



A Variational Approach to a Terrain Path Planning Problem with a Simulated Annealing Approximation

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Executive Summary

The Problem

In this dissertation, we explore path planning problems where the aim is to minimise cost. The cost of a path is made up of three components: terrain costs (i.e. hills), its curvature, and its length. We build upon the classical Dido problem to investigate this more complicated problem. A good analogy is planning a railway route or motorway, where infrastructure needs to be short, relatively straight, and cost-effective.

How the Issue Was Addressed

We start by applying the Classical Method of the Calculus of Variations. Immediately, we find that the Classical Method leads to a fourth-order ODE that we cannot solve explicitly. This motivates us to use the Direct Method of the Calculus of Variations, which provides an existence proof.

With the confidence of knowing that a minimiser exists, we apply a numerical stochastic optimisation method called Simulated Annealing, modelled in Python. This algorithm uses randomness to search for an approximation to an optimum. We start with an initial straight-line path and apply small movements. If the new path has a lower cost, we accept the move.

Interpretation of Findings

The existence proof is important, since it is not obvious that a minimising path exists. This dissertation also provides a high-level tutorial for learning how to use the Direct Method of the Calculus of Variations.

We had to make a few changes to the Simulated Annealing algorithm in order for it to work as intended. By changing how and where we applied these small movements, we were able to obtain interesting results, showing a path that moves around the hill while still being constrained by curvature and length.

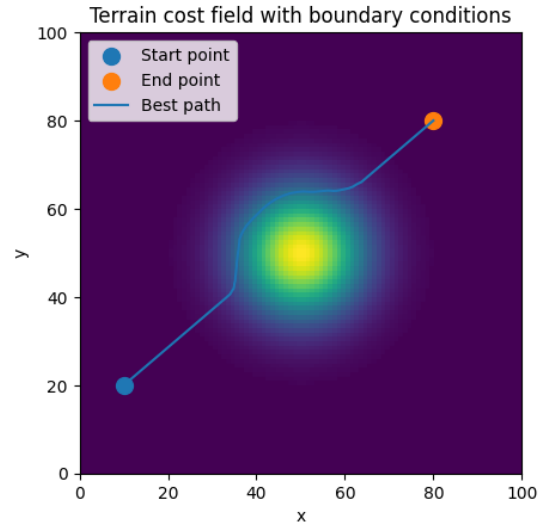


Figure 1: Final path obtained using simulated annealing.

Recommendations and Future Work

While the current model serves as a proof of concept, some avenues for further work are:

- **Real-World Data:** Use OpenStreetMap data to investigate whether the algorithm works effectively with real-world terrain and infrastructure. It would be particularly interesting to compare the routes produced by the algorithm with existing railway lines.
- **Three-Dimensional Extension:** Increase the complexity of the terrain cost function to take elevation into account, for example when considering tunnels or bridges. This would also allow factors such as the gradient of the track to be incorporated.
- **Systematic Parameter Tuning:** The parameters used in the numerical experiments were chosen through experimentation. A more systematic approach to selecting these parameters could be developed.

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Chapter 1

Introduction

The Calculus of Variations is one of the classical branches of mathematics. It was Euler who, looking at the work of Lagrange, gave the present name, not really self explanatory, to this field of mathematics.[4, p. 1]

In this dissertation, I explore applications of the Direct Method of the Calculus of Variations. The Calculus of Variations is concerned with optimisation problems where the quantity minimised or maximised depends upon a function. Although many variational problems can be stated in relatively simple terms, their solutions are often much more complex. Establishing the existence of an optimal function is not always straightforward. This can lead to non-linear differential equations that are difficult to solve.

The Classical Method of the Calculus of Variations is one approach to such problems. The conditions for a minimiser can be derived from the Euler–Lagrange equation. For fairly simple problems, this equation can be solved directly. However, as the functional becomes more complicated, the Euler–Lagrange equation may become a non-linear boundary-value problem that is difficult or impossible to obtain. This motivates the study of alternative approaches.

The main focus of this dissertation is the Direct Method of the Calculus of Variations. Rather than attempting to explicitly solve the Euler–Lagrange equation, the Direct Method seeks to establish the existence of a minimiser directly through the properties of the admissible space and the functional being minimised. The method relies upon ideas from functional analysis, including weak convergence, weak compactness, coercivity, and weak lower semicontinuity. A substantial part of the dissertation is devoted to developing the functional analysis background required to understand these ideas. We will be following Dacorogna’s *Introduction to the Calculus of Variations* [4].

The Classical Method is introduced through a number of simpler variational prob-

lems. Building upon the necessary results from functional analysis, the Direct Method is then developed and applied to establish the existence of minimisers for suitable variational problems.

The ideas developed throughout the dissertation are then applied to a path-planning problem. We want to plan routes on a 2D plane with two boundary points. We will be wanting to minimise the cost of the path along said plane, which is made up of the terrain cost, a curvature penalty and a length penalty. The resulting problem therefore seeks a route that will travel through lower-cost regions, avoiding unnecessary detours, and maintaining a degree of geometric smoothness.

The inclusion of this curvature term leads to a second-order variational problem and introduces analytical complexity. Although the Direct Method can be used to investigate the existence of minimisers, we do not get an explicit representation of the path. The corresponding Euler–Lagrange equation for this model leads to a non-linear fourth-order problem. For this reason, the final part of the dissertation considers the numerical approximation of optimal paths.

A stochastic optimisation method based on Simulated Annealing is developed to investigate this path-planning problem computationally. The continuous problem is discretised, allowing candidate paths to be represented numerically and iteratively modified in search of lower cost paths. The numerical experiments complement the theoretical analysis by searching for approximate solutions and illustrating the effect of the different terms within the functional.

The dissertation aims to connect the Direct Method with an optimisation problem and its numerical approximation. The progression from functional analysis, through the Classical and Direct Methods of the Calculus of Variations, to a numerical investigation of a path-planning problem reflects a central objective of this dissertation.

1.1 Background

The problems typically associated with the Calculus of Variations are ancient in comparison with many of the problems considered in modern mathematics, although their formulations can be simple to understand. At its core, the Calculus of Variations concerns optimisation problems in which one seeks to determine an optimal function with respect to an objective functional. Simple examples include finding the shortest path between two points in a plane or determining the maximum area enclosed by a fixed length of fencing.

Although the formulation of such problems may be relatively straightforward, the methods required to obtain their solutions can be considerably more involved. The de-

velopment of a systematic framework for the Calculus of Variations therefore allows more abstract and challenging problems to be considered. One such example is the problem of determining the optimal course of a sailing boat travelling against the wind, as discussed in Rindler's lecture notes, *Introduction to the Modern Calculus of Variations*. [7]

As in many works on the Calculus of Variations, I introduce the subject through Dido's problem. The semi-historical Phoenician princess, and later and most famously the founder and queen of Carthage, is said to have encountered a variational problem within the founding myth of her city. According to the legend, Dido was offered as much land as could be enclosed by the hide of a bull. She subsequently cut the hide into thin strips and used them to enclose a large area of land, which became the site of Carthage. In doing so, Dido is said to have encountered what is now known as the *isoperimetric problem*: *Find, among all curves of given length, the one which encloses the maximal area* [2].

It was this problem that inspired Euler to seek a general method for solving problems in the Calculus of Variations, laying some of the foundations of the subject. Eleven years after Euler's work, the nineteen-year-old Lagrange developed a more general method, which impressed Euler [2]. The Classical Method of the Calculus of Variations is therefore closely associated with the work of these two mathematical giants.

Following the development of the Classical Method, what is now known as the Direct Method was conceived through the work of mathematicians including Hilbert, Lebesgue, and Tonelli. Among the problems that motivated this transition was the Dirichlet integral: a problem of multiple integrals [4]. Hilbert played an important role in the transition of the Calculus of Variations into the modern era, particularly through the 19th, 20th, and 23rd problems that he presented at the International Congress of Mathematicians in Paris in 1900 [7].

This brings us to the present day, where the field of the Calculus of Variations is both broad and active. Mathematicians continue to develop the theory and apply it to an increasingly diverse range of problems, leaving their mark on the subject across areas of both pure and applied mathematics.

1.2 Problem Choice

It was initially hard for me to come to a decision regarding which problem to work on. The Calculus of Variations as a mathematical discipline has applications in a vast number of fields. After reading about Dido's problem and having a first discussion with my supervisor about possible problems, I was particularly attracted to path-planning problems.

At the start of my dissertation, I had begun reading about applications of optimisa-

tion and variational methods in economics and finance, including the work of Kamien and Schwartz [5] and Malliavin and Thalmaier [6]. This reading made clear that I lacked some of the analytical foundations required to pursue these problems rigorously. In particular, I understood the importance of developing a stronger functional-analytic foundations underlying the Direct Method of the Calculus of Variations. Following discussions with my supervisor, it was therefore decided that the dissertation would focus on developing these foundations, using Dacorogna's *Introduction to the Calculus of Variations* [4] as a key reference.

We considered a modified version of Dido's problem in which the objective is no longer to maximise area with a length constraint, but to determine an efficient path through a spatially varying cost field. The model includes penalties for both the length and curvature of the path. This formulation was motivated in part by a video from Wendover Productions [11] discussing the possible routes for the California High-Speed Rail project. The selection of a high-speed rail route must consider the different costs for each path, while also keeping journey length short and the geometric restrictions of high-speed rail.

The spatially varying cost field represents the differing costs of construction across a region, the length term penalises unnecessarily long routes, and the curvature term accounts for the geometric requirements of the railway. Although the motivating application is railway planning, the same mathematics could be applied to other large-scale infrastructure problems, such as motorway or pipeline route planning.

This problem was ultimately chosen because it provides both a meaningful application and a natural setting in which to investigate the Direct Method. The precise mathematical formulation and assumptions underlying the model are developed in Chapter 3, where we consider the associated minimisation problem including second order derivative form of the model (P).

Chapter 2

Mathematical Background

In this chapter, we introduce the foundational mathematics required for the study of the Calculus of Variations. The Classical Method leads to the Euler–Lagrange equation, which transforms a variational problem into an ordinary differential equation. Although the Classical Method is not the primary focus of this dissertation, an understanding of this approach provides useful motivation for the development of the Direct Method.

The Direct Method relies on several results from Functional Analysis, alongside an understanding of L^p and Sobolev spaces. Convexity is also an important tool in analysing variational problems and understanding the properties of their solutions. These concepts provide the mathematical framework required to establish the existence of minimisers using the Direct Method. Establishing existence before applying a numerical method is particularly useful, as it allows us to determine whether a variational problem admits a minimiser before attempting to approximate one computationally. Finally we will define and show some examples for the direct method, this will be used in the next chapter on our model problem.

We now formulate the general form of a Calculus of Variations problem. Throughout this dissertation, we focus primarily on minimisation problems. A general model problem can be written in the form

$$\inf \left\{ I(u) = \int_{\Omega} f(x, u, \nabla u(x)) \, dx : u \in X \right\} = m, \quad (\mathbf{P}_{\text{General}})$$

where the notation follows the formulation given by Dacorogna [4, p. 3].

Here, the various components of the problem are understood as follows:

- $\Omega \subset \mathbb{R}^n, n \geq 1$, is a bounded open set, with a point in Ω denoted by $x = (x_1, \dots, x_n)$;

- $u : \Omega \rightarrow \mathbb{R}^N$, $N \geq 1$, is the function being minimised, where

$$u = (u^1, \dots, u^N),$$

and

$$\nabla u = \left(\frac{\partial u^j}{\partial x_i} \right)_{\substack{1 \leq j \leq N \\ 1 \leq i \leq n}} \in \mathbb{R}^{N \times n};$$

- $f : \overline{\Omega} \times \mathbb{R}^N \times \mathbb{R}^{N \times n} \rightarrow \mathbb{R}$ is the integrand, assumed to be continuous;
- X denotes the space of admissible functions. For example, one may take $X = \{u \in C^1(\overline{\Omega}) : u = u_0 \text{ on } \partial\Omega\}$.

We are concerned with finding a *minimiser* $\bar{u} \in X$, that is, a function satisfying

$$I(\bar{u}) \leq I(u), \quad \forall u \in X.$$

All the mathematics we develop will be used with the goal of first proving the existence of minimisers and then searching for potential minimisers for these problems.

2.1 The Classical Method

We begin with a brief overview of the Classical Method, including some examples, in order to establish the ideas that motivate the Direct Method. Following the formulation in Dacorogna [4, p. 57], we consider the one-dimensional variational problem

$$\inf \{ I(u) : u \in C^1([a, b]), u(a) = \alpha, u(b) = \beta \}, \quad (\text{B})$$

where $f \in C^2([a, b] \times \mathbb{R} \times \mathbb{R})$ and

$$I(u) = \int_a^b f(x, u, u'(x)) dx.$$

To motivate the Classical Method, we first recall the analogous minimisation problem in $\mathbb{R}^n$,

$$\inf \{ F(x) : x \in X \subset \mathbb{R}^n \}.$$

For a sufficiently smooth minimiser $\bar{x}$, a necessary condition is

$$F'(\bar{x}) = 0.$$

The second and higher derivatives of F can then be used to investigate the nature of the critical point.

This formulation is a special case of the general variational problem introduced previously. In contrast to the general formulation, the Classical Method considers a one-dimensional domain $[a, b]$ and scalar-valued functions $u : [a, b] \rightarrow \mathbb{R}$. Consequently, the gradient ∇u is replaced by the ordinary derivative u' . The admissible functions are also restricted to $C^1([a, b])$ and are subject to fixed endpoint conditions $u(a) = \alpha$ and $u(b) = \beta$. The endpoint conditions will henceforth be referred to as boundary conditions.

To solve these problems, we can derive the Euler–Lagrange equation that should satisfy any $C^2([a, b])$ minimiser, $\bar{u}$, of (B). The Euler–Lagrange equation is analogous to $F'(\bar{x}) = 0$ in $\mathbb{R}^n$. [4]

The following theorem provides a necessary condition for a sufficiently smooth minimiser of the problem (B). It is stated here in the form used throughout this dissertation; the result corresponds to Theorem 2.1 in Dacorogna [4, Theorem 2.1, p. 59].

2.1.1 Euler–Lagrange equation

Theorem 2.1 (Euler–Lagrange equation). *Let $f \in C^2([a, b] \times \mathbb{R} \times \mathbb{R})$, $f = f(x, u, \xi)$, and*

$$\inf_{u \in X} \left\{ I(u) = \int_a^b f(x, u, u'(x)) dx \right\} = m, \quad (\mathbf{P}_{\text{Classical}})$$

where

$$X = \{u \in C^1([a, b]) : u(a) = \alpha, u(b) = \beta\}.$$

(i) *If $(\mathbf{P}_{\text{Classical}})$ admits a minimiser $\bar{u} \in X \cap C^2([a, b])$, then,*

$$\frac{d}{dx} [f_\xi(x, \bar{u}(x), \bar{u}'(x))] = f_u(x, \bar{u}(x), \bar{u}'(x)). \quad (\text{E})$$

(ii) *If $\bar{u}$ satisfies (E) and if $(u, \xi) \rightarrow f(x, u, \xi)$ is convex for every $x \in [a, b]$, then $\bar{u}$ is a minimiser of $(\mathbf{P}_{\text{Classical}})$.*

(iii) *If $(u, \xi) \rightarrow f(x, u, \xi)$ is **strictly** convex for every $x \in [a, b]$, then the minimiser of $(\mathbf{P}_{\text{Classical}})$, if it exists, is unique.*

Proof. (i) For every $h \in \mathbb{R}$ and every $v \in C^1([a, b])$ with $v(a) = v(b) = 0$ we have

$$I(\bar{u}) \leq I(\bar{u} + hv).$$

By setting $\Phi(h) = I(\bar{u} + hv)$, we have that $\Phi \in C^1(\mathbb{R})$ and $\Phi(0) \leq \Phi(h)$. We can deduce that Φ has a relative minimum at $h = 0$. Therefore,

$$\Phi'(0) = \left. \frac{d}{dh} [I(\bar{u} + hv)] \right|_{h=0} = 0.$$

Noting that

$$\left. \frac{d\Phi}{dh} \right|_{h=0} = \int_a^b \left. \frac{df}{dh} \right|_{h=0} dx,$$

differentiating $f(x, u, u')$ with respect to h , where $u = \bar{u} + hv$ and $u' = \bar{u}' + hv'$, gives

$$\begin{aligned} \frac{df}{dh} &= \frac{\partial f}{\partial x} \frac{dx}{dh} + \frac{\partial f}{\partial u} \frac{du}{dh} + \frac{\partial f}{\partial \xi} \frac{du'}{dh} \\ &= \frac{\partial f}{\partial u} v + \frac{\partial f}{\partial \xi} v', \end{aligned}$$

and hence

$$\Phi'(0) = \int_a^b [f_u(x, \bar{u}(x), \bar{u}'(x))v(x) + f_\xi(x, \bar{u}(x), \bar{u}'(x))v'(x)] dx = 0.$$

Using integration by parts on the second term, f_ξ , we get

$$\begin{aligned} \int_a^b f_\xi(x, \bar{u}(x), \bar{u}'(x))v'(x) dx &= \\ [f_\xi(x, \bar{u}(x), \bar{u}'(x))v(x)]_a^b - \int_a^b \left[\frac{d}{dx} f_\xi(x, \bar{u}(x), \bar{u}'(x)) \right] v(x) dx \end{aligned}$$

Since $v(a) = v(b) = 0$ the first term above vanishes. This leaves us with

$$\Phi'(0) = \int_a^b \left[f_u(x, \bar{u}(x), \bar{u}'(x)) - \frac{d}{dx} f_\xi(x, \bar{u}(x), \bar{u}'(x)) \right] v(x) dx = 0.$$

Applying the fundamental lemma of the calculus of variations, theorem 2.28, described in a further section, we obtain

$$f_u(x, \bar{u}(x), \bar{u}'(x)) - \frac{d}{dx} f_\xi(x, \bar{u}(x), \bar{u}'(x)) = 0$$

which is the Euler-Lagrange equation (E).

(ii) See Dacorogna [4, p. 61]

(iii) See Dacorogna [4, p. 62]

□

We can and will regularly write the Euler–Lagrange equation in its shorter form

$$f_y - \frac{d}{dx} f_\xi = 0.$$

Using Theorem 2.1, we can obtain minimisers for one-dimensional variational problems. We now consider several examples, starting with the problem of finding the shortest curve joining two points in the plane. Next, Dido’s problem before exploring a counterexample that shows the limitation of the Classical Method to be used as motivation for the development of the Direct Method.

Example 2.2. Find the shortest curve joining the points $A = (x_a, y_a)$ and $B = (x_b, y_b)$ in the plane, where $x_a < x_b$. We begin with the variational functional for arc length given as

$$J(y) = \int_{x_a}^{x_b} \sqrt{1 + (y'(x))^2} dx.$$

The integrand is

$$f(x, y, \xi) = \sqrt{1 + \xi^2},$$

which is independent of x and y . In particular, $f_y = 0$, so the Euler–Lagrange equation (E) reduces to

$$\frac{d}{dx} f_{y'}(x, y(x), y'(x)) = 0.$$

Differentiating f with respect to y' gives

$$f_{y'}(x, y(x), y'(x)) = \frac{y'(x)}{\sqrt{1 + (y'(x))^2}}. \quad (2.1)$$

Hence, by the reduced Euler–Lagrange equation, $f_{y'}$ is constant on $[x_a, x_b]$. Since (2.1) is a strictly increasing function of y' , it follows that y' is constant. Therefore,

$$y(x) = c_1 x + c_2,$$

for constants $c_1, c_2 \in \mathbb{R}$. Thus, the extremal is a straight line, as expected.

Example 2.3 (Dido’s problem). As discussed previously, Dido’s problem concerns finding a curve of prescribed length that encloses the maximum possible area. This problem can be formulated variationally as follows.

Among all sufficiently smooth functions y that are positive on $(0, \ell)$ and satisfy $y(0) = y(\ell) = 0$, find a function that minimises

$$J(y) = - \int_0^\ell y(x) \sqrt{1 - (y'(x))^2} dx.$$

The integrand is given as

$$f(x, y, \xi) = -y\sqrt{1 - \xi^2}.$$

Since f does not explicitly depend on x , we can use another reduced form of the Euler–Lagrange equation (E), so that

$$f - \xi f_\xi = \text{constant}. \quad (2.2)$$

Substituting our integrand into (2.2) gives us

$$\begin{aligned} f - y' f_\xi &\iff -y\sqrt{1 - y'^2} - y' \left(\frac{yy'}{\sqrt{1 - y'^2}} \right) = c \\ &\iff y\sqrt{1 - y'^2} + y' \left(\frac{yy'}{\sqrt{1 - y'^2}} \right) = \bar{c} \\ &\iff y(1 - y'^2) + yy'^2 = \bar{c}\sqrt{1 - y'^2} \\ &\iff y = \bar{c}\sqrt{1 - y'^2} \\ &\iff \bar{c}y' = \pm\sqrt{\bar{c}^2 - y^2} \quad \text{if } \bar{c} \neq 0, y = 0 \text{ otherwise.} \end{aligned}$$

We now have an ODE to solve. By separation of variables we obtain,

$$\int \frac{\bar{c}}{\sqrt{\bar{c}^2 - y^2}} dy = \pm \int dx.$$

Using the substitution, $y(u) = \bar{c} \sin(u) \implies dy = \bar{c} \cos(u) du$, on the left hand side, we get

$$\int \frac{\bar{c}}{\sqrt{\bar{c}^2 - y^2}} dy = \int \frac{\bar{c}^2 \cos(u)}{\bar{c} \cos(u)} du = \bar{c}u + \tilde{c}.$$

we therefore obtain

$$x = \bar{c}u + \tilde{c} \implies u = \frac{x}{\bar{c}} + \tilde{c}.$$

For $\bar{c} \neq 0$ we have

$$y(x) = \bar{c} \sin\left(\frac{x}{\bar{c}} + \tilde{c}\right).$$

as a solution to the variation problem and therefore an extremal. Using our boundary

conditions we can find a parametric representation as a solution to the problem.

$$(x(s), y(s)) = \left(-\frac{\ell}{\pi} \cos\left(\frac{\pi s}{\ell}\right) - \frac{\ell}{\pi}, \frac{\ell}{\pi} \sin\left(\frac{\pi s}{\ell}\right) \right) \quad (*)$$

the derivation of which is beyond the scope of this example.

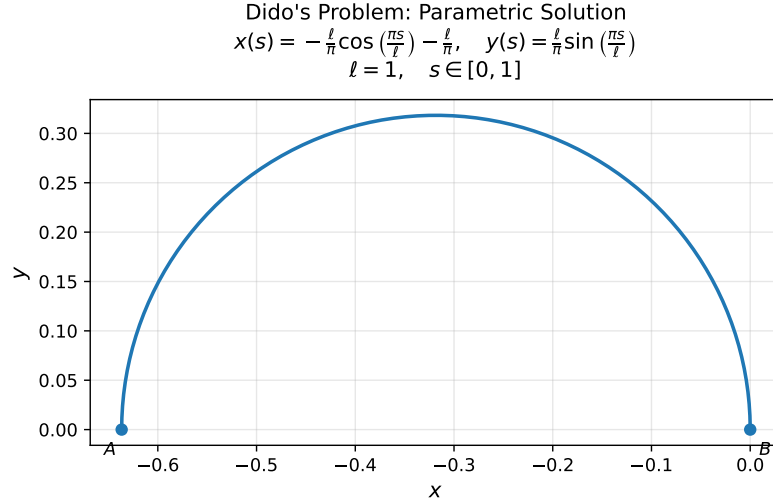


Figure 2.1: Parametric solution to Dido's problem given in (*) with $\ell = 1, s \in [0, 1]$.

For our counterexample, we consider the Lavrentiev phenomenon and, specifically, the Manià example [4, p. 70]. The Euler–Lagrange equation is derived under assumptions on the admissible functions, typically requiring them to be sufficiently smooth (C^1 or C^2). However, the Manià example demonstrates that a global minimiser of a variational problem need not possess the regularity required by the Classical Method. This motivates the need to consider more general function spaces and provides motivation for the development of the Direct Method.

Example 2.4 (Manià Example). Consider the variational problem

$$J(u) = \int_0^1 (x - u(x)^3)^2 (u'(x))^6 dx,$$

subject to the boundary conditions

$$u(0) = 0, \quad u(1) = 1.$$

The corresponding integrand is

$$f(x, u, \xi) = (x - u^3)^2 \xi^6.$$

Consider the function $\bar{u}(x) = x^{1/3}$ so that

$$J(\bar{u}(x)) = \int_0^1 (x - (x^{1/3})^3)^2 \left(\frac{1}{3x^{2/3}} \right)^6 = \int_0^1 0 \, dx = 0.$$

Since the integrand is non-negative, it follows that for every admissible function u

$$J(u) \geq 0.$$

Hence $\bar{u}$ is a global minimiser. However the derivative,

$$\bar{u}'(x) = \frac{1}{3}x^{-2/3},$$

becomes unbounded as $x \rightarrow 0$. Therefore $\bar{u} \notin C^1([0, 1])$. Thus, although $\bar{u}$ is a global minimiser, it does not possess the regularity assumed in the Classical Method. This demonstrates the need to consider broader classes of admissible functions and motivates the development of the Direct Method.

2.2 Continuous and Hölder Spaces

We introduce continuous and Hölder spaces following [4, Section 1.2, p. 14], selecting only those results relevant to this dissertation. Additional remarks and explanations are included where useful for developing the ideas required later.

2.2.1 Continuous Functions

Definition 2.5. Let $\Omega \subset \mathbb{R}^n$ be an open set.

- (ia) $C^0(\Omega) = C(\Omega)$ is the set of continuous **functions**, $u : \Omega \rightarrow \mathbb{R}$.
- (ib) $C^0(\Omega; \mathbb{R}^N) = C(\Omega; \mathbb{R}^N)$ is the set of continuous **maps**, $u : \Omega \rightarrow \mathbb{R}^N$.
- (ii) Let $C^0(\bar{\Omega}) = C(\bar{\Omega})$ be the set of continuous functions which can be **continuously extended** to $\bar{\Omega}$. Similar for maps.

(iii) The **support** of a function $u : \Omega \rightarrow \mathbb{R}$ is defined as

$$\text{supp } u = \overline{\{x \in \Omega : u(x) \neq 0\}}.$$

(iv) $C_0(\Omega) = \{u \in C(\Omega) : \text{supp } u \subset \Omega \text{ is compact}\}.$

(v) We define the norm over $C(\overline{\Omega})$ by,

$$\|u\|_{C^0} = \sup_{x \in \overline{\Omega}} |u(x)|.$$

Here are some examples to help build intuition for these definitions.

Example 2.6 (Support of a Function). The support of a function is the closure of the region where the function is *not zero*. Let

$$u(x) = \begin{cases} 1 - x^2, & \text{if } x \in (-1, 1), \\ 0, & \text{otherwise.} \end{cases}$$

Then:

- $u(x) \neq 0$ only for $x \in (-1, 1)$;
- therefore,

$$\{x \in \Omega : u(x) \neq 0\} = (-1, 1);$$

- hence, the support is

$$\text{supp}(u) = \overline{(-1, 1)} = [-1, 1].$$

Remark 2.7. $C(\overline{\Omega})$ equipped with the norm $\|\cdot\|_{C^0}$ is Banach space

Theorem 2.8 (Ascoli–Arzelà theorem). Let $\Omega \subset \mathbb{R}^n$ be a bounded domain. Let $K \subset C(\overline{\Omega})$ be bounded such that the following property of equicontinuity holds: $\forall \varepsilon > 0$, there exists $\delta > 0$

$$|x - y| < \delta \implies |u(x) - u(y)| < \varepsilon, \forall x, y \in \overline{\Omega} \text{ and } \forall u \in K,$$

Then $\overline{K}$ is compact.

Example 2.9 (Equicontinuity and Ascoli–Arzelà). Consider the sequence of functions on $[0, 1]$ given by

$$u_n(x) = \frac{x}{n}.$$

Each u_n is continuous on $[0, 1]$, and

$$0 \leq u_n(x) \leq 1,$$

for all $x \in [0, 1]$ and $n \geq 1$. Furthermore, for any $x_1, x_2 \in [0, 1]$,

$$|u_n(x_1) - u_n(x_2)| = \frac{1}{n} |x_1 - x_2| \leq |x_1 - x_2|.$$

Thus, the family $\{u_n\}$ is equicontinuous, since the same bound controls the change in every function in the family.

The Ascoli–Arzelà theorem (2.8) therefore guarantees that the sequence contains a uniformly convergent subsequence. In fact, in this example the whole sequence converges uniformly to the zero function:

$$\|u_n\|_{C^0} = \sup_{x \in [0,1]} \left| \frac{x}{n} \right| = \frac{1}{n} \longrightarrow 0.$$

Remark 2.10. It is important that the hypotheses of the Ascoli–Arzelà theorem (2.8) are satisfied. For example, consider

$$u_n(x) = x^n, \quad x \in [0, 1].$$

Although each u_n is continuous and the sequence is uniformly bounded, the family $\{u_n\}$ is not equicontinuous on $[0, 1]$. Indeed, the functions become increasingly steep near $x = 1$.

Pointwise, we have

$$u_n(x) \longrightarrow \begin{cases} 0, & 0 \leq x < 1, \\ 1, & x = 1. \end{cases}$$

whose limit is not continuous. Consequently, the sequence cannot have a uniformly convergent subsequence.

Remark 2.11. We introduce the notation for multi-index derivatives, which may not be familiar. Let

$$a = (a_1, \dots, a_n),$$

be a multi-index, where each a_i is a non-negative integer. The order of a is defined by

$$|a| = a_1 + \dots + a_n.$$

For a sufficiently differentiable function $u : \Omega \rightarrow \mathbb{R}$, the derivative corresponding to the

multi-index a is

$$D^a u = \frac{\partial^{|a|} u}{\partial x_1^{a_1} \cdots \partial x_n^{a_n}}.$$

In other words, a_i tells us how many times to differentiate with respect to x_i .

Definition 2.12. Let $\Omega \subset \mathbb{R}^n$ be an open set and $k \geq 0$ be an integer.

- (i) Set of continuous Functions, $u : \Omega \rightarrow \mathbb{R}$, which have all partial derivatives, $D^a u$, $a \in \mathcal{A}_m$, $0 \leq m \leq k$, is $C^k(\Omega)$.
- (ii) $C^k(\overline{\Omega})$ is the set of functions whose derivatives up to the order k can be extended continuously to $\overline{\Omega}$. It is equipped with the following norm,

$$\|u\|_{C^k} = \max_{0 \leq |a| \leq k} \sup_{x \in \overline{\Omega}} |D^a u(x)|.$$

- (iii) $C_0^k(\Omega) = C^k(\Omega) \cap C_0(\Omega)$.
- (iv) $C^\infty(\Omega) = \cap_{k=0}^\infty C^k(\Omega)$.
- (v) $C_0^\infty(\Omega) = C^\infty(\Omega) \cap C_0(\Omega)$.

2.2.2 Hölder Continuous Functions

We now turn to the notion of Hölder continuous functions.

Definition 2.13 (Hölder Seminorm). Let $D \subset \mathbb{R}^n$, $u : D \rightarrow \mathbb{R}$ and $0 < \alpha \leq 1$. The Hölder seminorm of u is defined by

$$[u]_{C^{0,\alpha}(D)} = \sup_{\substack{x,y \in D \\ x \neq y}} \frac{|u(x) - u(y)|}{|x - y|^\alpha}.$$

Definition 2.14 (Hölder Spaces). Let $D \subset \mathbb{R}^n$, $u : D \rightarrow \mathbb{R}$ and $0 < \alpha \leq 1$.

- (i) $C^{0,\alpha}(\Omega)$ is the set of functions $u \in C(\Omega)$ such that

$$[u]_{C^{0,\alpha}(K)} < \infty \quad \text{for every compact set } K \subset \Omega.$$

- (ii) $C^{0,\alpha}(\overline{\Omega})$ is the set of functions $u \in C(\overline{\Omega})$ such that

$$[u]_{C^{0,\alpha}(\overline{\Omega})} < \infty.$$

This space is equipped with the norm

$$\|u\|_{C^{0,\alpha}(\overline{\Omega})} = \|u\|_{C^0(\overline{\Omega})} + [u]_{C^{0,\alpha}(\overline{\Omega})}.$$

(iii) For $k \in \mathbb{N}$, let

$$\mathcal{A}_k = \{a \in \mathbb{N}_0^n : |a| \leq k\}.$$

Then $C^{k,\alpha}(\Omega)$ is the set of functions $u \in C^k(\Omega)$ such that

$$[D^a u]_{C^{0,\alpha}(K)} < \infty \quad \text{for every compact set } K \subset \Omega \text{ and every } a \in \mathcal{A}_k.$$

(iv) Similarly, $C^{k,\alpha}(\overline{\Omega})$ is the set of functions $u \in C^k(\overline{\Omega})$ such that

$$[D^a u]_{C^{0,\alpha}(\overline{\Omega})} < \infty \quad \text{for every } a \in \mathcal{A}_k.$$

This space is equipped with the norm

$$\|u\|_{C^{k,\alpha}(\overline{\Omega})} = \|u\|_{C^k(\overline{\Omega})} + \max_{a \in \mathcal{A}_k} [D^a u]_{C^{0,\alpha}(\overline{\Omega})}.$$

Remark 2.15 (Lipschitz Continuity). When $\alpha = 1$, the space $C^{0,1}(\overline{\Omega})$ is the space of *Lipschitz continuous functions*. In particular, a function $u \in C(\overline{\Omega})$ is Lipschitz continuous if there exists a constant $\gamma > 0$ such that

$$|u(x) - u(y)| \leq \gamma |x - y|, \quad \forall x, y \in \overline{\Omega}.$$

The smallest possible value of γ is given by the Hölder seminorm

$$\gamma = [u]_{C^{0,1}(\overline{\Omega})}.$$

Example 2.16 (Hölder Continuous Function and Computation of the Norm). Let $D = [0, 1]$ and define

$$u(x) = \sqrt{x}.$$

We compute the Hölder norm of u in $C^{0,\frac{1}{2}}([0, 1])$.

First, the C^0 norm is

$$\|u\|_{C^0} = \sup_{x \in [0,1]} |\sqrt{x}| = 1.$$

Next, taking $\alpha = \frac{1}{2}$, the Hölder seminorm is

$$[u]_{C^{0,\frac{1}{2}}} = \sup_{\substack{x,y \in [0,1] \\ x \neq y}} \frac{|\sqrt{x} - \sqrt{y}|}{|x - y|^{1/2}}.$$

Using the identity

$$|\sqrt{x} - \sqrt{y}| = \frac{|x - y|}{\sqrt{x} + \sqrt{y}},$$

we obtain

$$\frac{|\sqrt{x} - \sqrt{y}|}{|x - y|^{1/2}} = \frac{|x - y|^{1/2}}{\sqrt{x} + \sqrt{y}}.$$

Without loss of generality, suppose that $x > y$. Then

$$\frac{|x - y|^{1/2}}{\sqrt{x} + \sqrt{y}} = \frac{\sqrt{x - y}}{\sqrt{x} + \sqrt{y}} \leq 1.$$

Equality is attained when $y = 0$, giving

$$[u]_{C^{0,\frac{1}{2}}} = 1.$$

Therefore, the full Hölder norm is

$$\|u\|_{C^{0,\frac{1}{2}}} = \|u\|_{C^0} + [u]_{C^{0,\frac{1}{2}}} = 1 + 1 = 2.$$

This example illustrates that $u(x) = \sqrt{x}$ is continuous but not Lipschitz continuous.

Example 2.17 ($C^{1,\alpha}$ Norm in $\mathbb{R}^2$). Let $\Omega = [0, 1]^2$ and define

$$u(x, y) = x^2 + y^2.$$

We illustrate the components of the $C^{1,\alpha}(\Omega)$ norm.

First, we compute the first partial derivatives:

$$u_x = 2x, \quad u_y = 2y.$$

The C^1 norm consists of the supremum norms of the function and its first derivatives. We have

$$\sup_{\Omega} |u| = 2, \quad \sup_{\Omega} |u_x| = 2, \quad \sup_{\Omega} |u_y| = 2.$$

Therefore,

$$\|u\|_{C^1} = \sup_{\Omega} |u| + \sup_{\Omega} |u_x| + \sup_{\Omega} |u_y| = 2 + 2 + 2 = 6.$$

Now consider the Hölder seminorms of the first derivatives. For $(x, y), (x', y') \in \Omega$,

$$|u_x(x, y) - u_x(x', y')| = 2|x - x'| \leq 2\|(x, y) - (x', y')\|,$$

and similarly,

$$|u_y(x, y) - u_y(x', y')| = 2|y - y'| \leq 2\|(x, y) - (x', y')\|.$$

In particular, when $\alpha = 1$, we have

$$[u_x]_{C^{0,1}(\Omega)} = [u_y]_{C^{0,1}(\Omega)} = 2.$$

Hence,

$$\max_{|a|=1} [D^a u]_{C^{0,1}(\Omega)} = 2.$$

Therefore, using the definition of the $C^{1,1}$ norm,

$$\|u\|_{C^{1,1}(\Omega)} = \|u\|_{C^1(\Omega)} + \max_{|a| \leq 1} [D^a u]_{C^{0,1}(\Omega)}$$

and, with the convention used here,

$$\|u\|_{C^{1,1}(\Omega)} = 6 + 2 = 8.$$

2.3 Lebesgue Spaces

In this section, we introduce Lebesgue spaces, commonly denoted by L^p spaces. These spaces allow us to work with weaker notions of regularity than those required by the Classical Method, by focusing on the integrability of functions rather than their pointwise smoothness. They are also essential for the definition of Sobolev spaces, which will be introduced in the following section. This provides the language and tools needed to consider whether a function, or its derivatives, are integrable, rather than requiring the stronger pointwise regularity imposed by the Classical Method.

Definition 2.18. Let $\Omega \subset \mathbb{R}^n$ be an open set and $1 \leq p \leq \infty$. We say that a measurable

function $u : \Omega \rightarrow \mathbb{R}$ belongs to $L^p(\Omega)$ if $\|u\|_{L^p}$ is finite. For p finite

$$\|u\|_{L^p} = \left(\int_{\Omega} |u(x)|^p dx \right)^{\frac{1}{p}},$$

and for $p = \infty$

$$\|u\|_{L^p} = \inf\{\alpha : |u(x)| < \alpha \text{ a.e. in } \Omega\}.$$

Example 2.19 (Example of L^p Norms). Let $\Omega = (0, 1)$ and define

$$u(x) = x.$$

(i) **Case $p < \infty$.**

Consider $p = 2$. Then

$$\|u\|_{L^2(0,1)} = \left(\int_0^1 |x|^2 dx \right)^{1/2} = \left(\frac{1}{3} \right)^{1/2} = \frac{1}{\sqrt{3}}.$$

Since the norm is finite, it follows that

$$u \in L^2(0, 1).$$

(ii) **Case $p = \infty$.**

The function $u(x) = x$ takes values arbitrarily close to 1 on $(0, 1)$, but does not attain the value 1. Its essential supremum is therefore

$$\|u\|_{L^\infty(0,1)} = \operatorname{ess\,sup}_{x \in (0,1)} |u(x)| = 1.$$

Since the essential supremum is finite, it follows that

$$u \in L^\infty(0, 1).$$

Example 2.20 (A Function in L^1 but not in L^2). Let

$$u(x) = \frac{1}{\sqrt{x}},$$

defined on $\Omega = (0, 1)$. First, consider the L^1 norm. We have

$$\|u\|_{L^1(0,1)} = \int_0^1 |u(x)| dx = \int_0^1 x^{-1/2} dx = 2.$$

Hence,

$$u \in L^1(0, 1).$$

However, for the L^2 norm,

$$\|u\|_{L^2(0,1)}^2 = \left(\int_0^1 |u(x)|^2 dx \right)^{\frac{1}{2}} = \left(\int_0^1 x^{-1} dx \right)^{\frac{1}{2}} = \infty.$$

Therefore,

$$u \notin L^2(0, 1).$$

This example illustrates that a function may be integrable in the L^1 sense without being square-integrable. In particular, membership of a function in one L^p space does not automatically imply membership in another.

Remark 2.21. In the following theorems, we will make use of the *conjugate exponent* of p , denoted by p' . For $1 < p < \infty$, it is defined by

$$\frac{1}{p} + \frac{1}{p'} = 1, \quad \text{equivalently} \quad p' = \frac{p}{p-1}.$$

For the endpoint cases, if $p = 1$, we define $p' = \infty$, while if $p = \infty$, we define $p' = 1$.

Theorem 2.22 (Basic properties of L^p spaces). *Let $\Omega \subset \mathbb{R}^n$ be an open set and let $1 \leq p \leq \infty$. Then the following hold.*

- (i) *The space $L^p(\Omega)$ is a Banach space. Moreover, $L^2(\Omega)$ is a **Hilbert space** with inner product*

$$\langle u, v \rangle = \int_{\Omega} u(x)v(x) dx.$$

- (ii) **Hölder's inequality:** *If $u \in L^p(\Omega)$ and $v \in L^{p'}(\Omega)$, where p' is the conjugate exponent of p , then $uv \in L^1(\Omega)$ and*

$$\|uv\|_{L^1(\Omega)} \leq \|u\|_{L^p(\Omega)} \|v\|_{L^{p'}(\Omega)}.$$

*In particular, when $p = p' = 2$, this gives the **Cauchy–Schwarz inequality***

$$\|uv\|_{L^1(\Omega)} \leq \|u\|_{L^2(\Omega)} \|v\|_{L^2(\Omega)}.$$

(iii) **Minkowski's inequality:** If $u, v \in L^p(\Omega)$, then

$$\|u + v\|_{L^p(\Omega)} \leq \|u\|_{L^p(\Omega)} + \|v\|_{L^p(\Omega)}.$$

(iv) **Riesz theorem:** If $1 \leq p < \infty$ and $\varphi \in (L^p(\Omega))'$, then, under the appropriate assumptions, there exists $u \in L^{p'}(\Omega)$ such that

$$\langle \varphi, f \rangle = \varphi(f) = \int_{\Omega} u(x)f(x) dx, \quad \forall f \in L^p(\Omega).$$

and moreover

$$\|u\|_{L^{p'}} = \|\varphi\|_{(L^p)'}.$$

(v) The space $L^p(\Omega)$ is separable for $1 \leq p < \infty$ and reflexive for $1 < p < \infty$.

(vi) If $1 \leq p < \infty$ and $u \in L^p(\Omega)$, then there exists a sequence $(u_\nu) \subset C_0^\infty(\Omega)$ such that

$$\lim_{\nu \rightarrow \infty} \|u_\nu - u\|_{L^p(\Omega)} = 0.$$

This result is false in general for $p = \infty$.

Remark 2.23 (Duality of L^p Spaces). To summarise the duality results, we have

$$(L^p(\Omega))' \cong L^{p'}(\Omega), \quad 1 < p < \infty,$$

where p' is the conjugate exponent of p . In particular,

$$(L^2(\Omega))' \cong L^2(\Omega),$$

since $2' = 2$. At the endpoint $p = 1$, we have

$$(L^1(\Omega))' \cong L^\infty(\Omega).$$

However, the corresponding result does not hold for $p = \infty$. In general,

$$L^1(\Omega) \subsetneq (L^\infty(\Omega))'.$$

We will now consider convergence in L^p spaces. The natural notion of *strong convergence* is induced by the L^p norm. We will also introduce the notion of *weak convergence*, which will be important when considering the Direct Method.

Definition 2.24 (Convergence in L^p spaces). Let $\Omega \subset \mathbb{R}^n$ be an open set and let $1 \leq p \leq \infty$.

- (i) A sequence (u_ν) is said to **strongly converge** to u in $L^p(\Omega)$ if $u_\nu, u \in L^p(\Omega)$ and

$$\lim_{\nu \rightarrow \infty} \|u_\nu - u\|_{L^p(\Omega)} = 0.$$

We denote this by

$$u_\nu \rightarrow u \quad \text{in } L^p(\Omega).$$

- (ii) If $1 \leq p < \infty$, a sequence (u_ν) is said to **weakly converge** to u in $L^p(\Omega)$ if $u_\nu, u \in L^p(\Omega)$ and

$$\lim_{\nu \rightarrow \infty} \int_{\Omega} [u_\nu(x) - u(x)] \varphi(x) \, dx = 0, \quad \forall \varphi \in L^{p'}(\Omega).$$

We denote this by

$$u_\nu \rightharpoonup u \quad \text{in } L^p(\Omega).$$

- (iii) If $p = \infty$, a sequence (u_ν) is said to **weak-* converge** to u in $L^\infty(\Omega)$ if $u_\nu, u \in L^\infty(\Omega)$ and

$$\lim_{\nu \rightarrow \infty} \int_{\Omega} [u_\nu(x) - u(x)] \varphi(x) \, dx = 0, \quad \forall \varphi \in L^1(\Omega).$$

We denote this by

$$u_\nu \rightharpoonup^* u \quad \text{in } L^\infty(\Omega).$$

Example 2.25 (Weak Convergence in L^p). Let $\Omega = (0, 1)$ and define

$$u_\nu(x) = \sin(2\pi\nu x).$$

We claim that

$$u_\nu \rightharpoonup 0 \quad \text{in } L^p(0, 1), \quad 1 < p < \infty.$$

Indeed, let $\varphi \in L^{p'}(0, 1)$. Then

$$\int_0^1 u_\nu(x) \varphi(x) \, dx = \int_0^1 \sin(2\pi\nu x) \varphi(x) \, dx \longrightarrow 0 \quad \text{as } \nu \rightarrow \infty.$$

This follows from the oscillatory nature of $\sin(2\pi\nu x)$ and the Riemann–Lebesgue lemma.

Therefore,

$$u_\nu \rightharpoonup 0 \quad \text{in } L^p(0, 1).$$

However, the convergence is not strong. For example, when $p = 2$,

$$\|u_\nu\|_{L^2(0,1)}^2 = \int_0^1 \sin^2(2\pi\nu x) \, dx = \frac{1}{2},$$

and hence

$$\|u_\nu\|_{L^2(0,1)} = \frac{1}{\sqrt{2}} \not\rightarrow 0.$$

Thus, weak convergence does not in general imply strong convergence.

Example 2.26 (Weak-* Convergence in L^∞). Let $\Omega = (0, 1)$ and define the sequence (u_ν) by

$$u_\nu(x) = (-1)^k, \quad x \in \left[\frac{k}{\nu}, \frac{k+1}{\nu}\right), \quad k = 0, \dots, \nu - 1.$$

Thus, u_ν alternates between 1 and -1 on increasingly fine subintervals of $(0, 1)$. In particular,

$$\|u_\nu\|_{L^\infty(0,1)} = 1,$$

so the sequence is bounded in $L^\infty(0, 1)$.

For every $\varphi \in L^1(0, 1)$, the increasingly rapid oscillations of u_ν imply

$$\int_0^1 u_\nu(x) \varphi(x) \, dx \longrightarrow 0 \quad \text{as } \nu \rightarrow \infty.$$

Hence,

$$u_\nu \rightharpoonup^* 0 \quad \text{in } L^\infty(0, 1).$$

However,

$$\|u_\nu - 0\|_{L^\infty(0,1)} = 1$$

for every ν . Therefore, u_ν does not converge strongly to zero in $L^\infty(0, 1)$.

This example illustrates that weak-* convergence does not in general imply strong convergence in L^∞ .

Definition 2.27. $u \in L_{loc}^p(\Omega)$ if $u \in L^p(\Omega')$ for every open set Ω' compactly contained in Ω

Theorem 2.28 (Fundamental lemma of the calculus of variations). *Let Ω be an open set*

of $\mathbb{R}^n$ and $u \in L^1_{loc}(\Omega)$ be such that

$$\int_{\Omega} u(x) \varphi(x) \, dx = 0, \quad \forall \varphi \in C_0^\infty(\Omega)$$

then $u = 0$, a.e. in Ω .

Proof. Following the method presented by Dacorogna [4, p. 27], we first prove the theorem under the stronger assumption that $u \in L^2(\Omega)$. Recall that $L^2(\Omega) \subset L^1_{loc}(\Omega)$. Let $\epsilon > 0$. By part (iv) of the Riesz theorem, stated in Theorem 2.22, and since $u \in L^2(\Omega)$, there exists $\varphi \in C_0^\infty(\Omega)$ such that

$$\|u - \varphi\|_{L^2} \leq \epsilon.$$

By the hypothesis of the fundamental lemma,

$$\int_{\Omega} u \varphi \, dx = 0.$$

Therefore,

$$\begin{aligned} \|u\|_{L^2}^2 &= \int_{\Omega} u^2 \, dx \\ &= \int_{\Omega} u^2 \, dx - \int_{\Omega} u \varphi \, dx \\ &= \int_{\Omega} u(u - \varphi) \, dx. \end{aligned}$$

Applying Hölder's inequality, given in part (ii) of Theorem 2.22, we obtain

$$\|u\|_{L^2}^2 \leq \|u\|_{L^2} \|u - \varphi\|_{L^2} \leq \epsilon \|u\|_{L^2}.$$

Since $\epsilon > 0$ is arbitrary, it follows that

$$\|u\|_{L^2} = 0.$$

Hence $u = 0$ almost everywhere in Ω , as required. $\square$

2.4 Sobolev Spaces

In this section, we introduce Sobolev spaces, commonly denoted by $W^{k,p}(\Omega)$, where $1 \leq p \leq \infty$ and $k \in \mathbb{N}$. We weaken the notion of differentiation so that we can work with functions that are not necessarily classically differentiable, while retaining the ability

to integrate by parts. This is particularly useful in the Direct Method, where we need to control both a function and its derivatives. We begin by introducing weak derivatives before defining Sobolev spaces and giving some examples for intuition. For further reading, see Dacorogna [4, p. 30].

Weak Partial Derivative

Definition 2.29. Let $\Omega \subset \mathbb{R}^N$ be open and $u \in L^1_{loc}(\Omega)$. We say that:

- (i) $v \in L^1_{loc}(\Omega)$ is the *weak partial derivative* of u with respect to x_i if

$$\int_{\Omega} v(x) \varphi(x) dx = - \int_{\Omega} u(x) \frac{\partial \varphi}{\partial x_i} dx, \quad \forall \varphi \in C_0^\infty(\Omega).$$

by abuse of notation, $v = \delta u / \delta x_i$ or u_{x_i} .

- (ii) We say that u is *weakly differentiable* if the weak partial derivatives, $u_{x_i}, \dots, u_{x_n}$, exist.

Example 2.30 (Weak derivative example). Let us consider the function

$$u(x) = |x|, \quad x \in (-1, 1).$$

This function is not classically differentiable at $x = 0$. However, we can show that it possesses a weak derivative. Define

$$v(x) = \begin{cases} -1, & x < 0, \\ 1, & x > 0, \end{cases}$$

or, equivalently,

$$v(x) = \operatorname{sgn}(x).$$

We want to show that v is the weak derivative of u . From the definition of a weak derivative, we require

$$\int_{-1}^1 v(x) \varphi(x) dx = - \int_{-1}^1 |x| \varphi'(x) dx, \quad \forall \varphi \in C_0^\infty(-1, 1).$$

Splitting the integral at $x = 0$ gives

$$- \int_{-1}^1 |x| \varphi'(x) dx = \int_{-1}^0 x \varphi'(x) dx - \int_0^1 x \varphi'(x) dx.$$

Integrating by parts,

$$\int_{-1}^0 x\varphi'(x) dx = [x\varphi(x)]_{-1}^0 - \int_{-1}^0 \varphi(x) dx = - \int_{-1}^0 \varphi(x) dx,$$

where the boundary terms vanish since $\varphi \in C_0^\infty(-1, 1)$. Similarly,

$$- \int_0^1 x\varphi'(x) dx = \int_0^1 \varphi(x) dx.$$

Hence,

$$- \int_{-1}^1 |x|\varphi'(x) dx = - \int_{-1}^0 \varphi(x) dx + \int_0^1 \varphi(x) dx = \int_{-1}^1 \operatorname{sgn}(x)\varphi(x) dx.$$

Therefore,

$$u_x = \operatorname{sgn}(x)$$

is the weak derivative of u .

We will now consider Sobolev spaces with weak derivatives of order one, denoted by $W^{1,p}(\Omega)$.

2.4.1 Sobolev Norms

Definition 2.31. Let $\Omega \subset \mathbb{R}^n$ be an open set and $1 \leq p \leq \infty$.

- (i) We let $W^{1,p}(\Omega)$ be the set of functions $u : \Omega \rightarrow \mathbb{R}^n$, $u \in L^p(\Omega)$, whose weak partial derivatives $u_{x_i} \in L^p(\Omega)$ for every $i = 1, \dots, n$. This space has the following norm,

$$\|u\|_{W^{1,p}(\Omega)} = \begin{cases} \left(\|u\|_{L^p(\Omega)}^p + \|\nabla u\|_{L^p(\Omega)}^p \right)^{1/p}, & 1 \leq p < \infty, \\ \max \{ \|u\|_{L^\infty(\Omega)}, \|\nabla u\|_{L^\infty(\Omega)} \}, & p = \infty. \end{cases}$$

In the special case of $p = 2$, the space $W^{1,2}(\Omega)$ is commonly denoted by $H^1(\Omega)$, as it forms a Hilbert space with its natural inner product;

- (ii) For $1 \leq p < \infty$, the set $W_0^{1,p}(\Omega)$ is the closure of $C_0^\infty(\Omega)$ functions in $W^{1,p}(\Omega)$. If Ω is bounded then $u \in W_0^{1,p}(\Omega)$ is such that $u \in W^{1,p}(\Omega)$ and $u = 0$ on $\partial\Omega$;
- (iii) $u \in u_0 + W_0^{1,p}(\Omega)$ means $u, u_0 \in W^{1,p}(\Omega)$ and $u - u_0 \in W_0^{1,p}(\Omega)$;
- (iv) We let $W_0^{1,\infty} = W^{1,\infty} \cap W_0^{1,1}$.

Example 2.32 (Computing the Sobolev norm). Let us again consider the function

$$u(x) = |x|, \quad x \in (-1, 1).$$

From Example 2.30, we have

$$u_x = \operatorname{sgn}(x).$$

For $p = 1$, from part (i) of Definition 2.31,

$$\|u\|_{W^{1,1}} = \|u\|_{L^1} + \|u_x\|_{L^1}.$$

We compute each term separately:

$$\|u\|_{L^1} = \int_{-1}^1 |x| dx = 2 \int_0^1 x dx = 1,$$

and

$$\|u_x\|_{L^1} = \int_{-1}^1 |\operatorname{sgn}(x)| dx = \int_{-1}^1 1 dx = 2.$$

Therefore,

$$\|u\|_{W^{1,1}} = 3.$$

For $p = 2$,

$$\|u\|_{W^{1,2}} = (\|u\|_{L^2}^2 + \|u_x\|_{L^2}^2)^{1/2}.$$

We have

$$\|u\|_{L^2}^2 = \int_{-1}^1 |x|^2 dx = \frac{2}{3},$$

and

$$\|u_x\|_{L^2}^2 = \int_{-1}^1 |\operatorname{sgn}(x)|^2 dx = \int_{-1}^1 1 dx = 2.$$

Hence,

$$\|u\|_{W^{1,2}} = \sqrt{\frac{2}{3} + 2} = \sqrt{\frac{8}{3}}.$$

Finally, for $p = \infty$,

$$\|u\|_{W^{1,\infty}} = \max \{ \|u\|_{L^\infty}, \|u_x\|_{L^\infty} \}.$$

Since

$$\|u\|_{L^\infty} = \sup_{x \in (-1,1)} |x| = 1,$$

and

$$\|u_x\|_{L^\infty} = \sup_{x \in (-1,1)} |\operatorname{sgn}(x)| = 1,$$

we obtain

$$\|u\|_{W^{1,\infty}} = 1.$$

We now consider Sobolev spaces for $k \neq 1$.

Definition 2.33. Let $\Omega \subset \mathbb{R}^n$ be an open set and $1 \leq p \leq \infty$.

- (i) For higher-order derivatives, we define the Sobolev spaces as follows. For $k \in \mathbb{N}$, $k \geq 1$, the space $W^{k,p}(\Omega)$ consists of all functions $u \in L^p(\Omega)$ whose weak partial derivatives $D^\alpha u$ belong to $L^p(\Omega)$ for every multi-index α satisfying $0 \leq |\alpha| \leq k$. The corresponding norm is defined by

$$\|u\|_{W^{k,p}(\Omega)} = \begin{cases} \left(\sum_{|\alpha| \leq k} \|D^\alpha u\|_{L^p(\Omega)}^p \right)^{1/p}, & 1 \leq p < \infty, \\ \max_{|\alpha| \leq k} \|D^\alpha u\|_{L^\infty(\Omega)}, & p = \infty; \end{cases}$$

- (ii) If $1 \leq p < \infty$, the set $W_0^{k,p}(\Omega)$ is the closure of $C_0^\infty(\Omega)$ in $W^{k,p}(\Omega)$ and $W_0^{k,p}(\Omega) = W^{k,\infty}(\Omega) \cap W_0^{k,1}(\Omega)$.

Having defined $W^{1,p}(\Omega)$, we can interpret k as the largest order of weak derivative controlled by the space. Thus, increasing k requires progressively higher-order regularity:

1. $k = 0$: control of the function itself through integrability;
2. $k = 1$: control of first-order weak derivatives;
3. $k = 2$: control of second-order weak derivatives;
4. higher k : control of higher-order weak derivatives.

p , on the other hand, determines the integrability of the function and its derivatives. Roughly speaking, it determines how strongly large values are penalised in the corresponding L^p norm. Smaller values of p allow greater variation, while larger values place greater emphasis on large values. In the case of $p = \infty$, the L^∞ norm essentially measures the maximum value.

Therefore, k determines *what order of derivative* is controlled, while p determines *how the size of those derivatives is measured*.

Example 2.34 (Sobolev norm for general k). Let us consider the function

$$u(x) = x^2, \quad x \in (-1, 1).$$

We compute the $W^{2,2}$ norm for this function. We have

$$\|u\|_{W^{2,2}}^2 = \|u\|_{L^2}^2 + \|u'\|_{L^2}^2 + \|u''\|_{L^2}^2.$$

Our derivatives are

$$u(x) = x^2, \quad u'(x) = 2x, \quad u''(x) = 2.$$

Using the definition of the L^2 norm, we have

$$\|u\|_{L^2}^2 = \int_{-1}^1 x^4 dx = \frac{2}{5},$$

$$\|u'\|_{L^2}^2 = \int_{-1}^1 (2x)^2 dx = \int_{-1}^1 4x^2 dx = \frac{8}{3},$$

and

$$\|u''\|_{L^2}^2 = \int_{-1}^1 4 dx = 8.$$

Therefore,

$$\|u\|_{W^{2,2}}^2 = \frac{2}{5} + \frac{8}{3} + 8 = \frac{166}{15}.$$

Hence,

$$\|u\|_{W^{2,2}} = \sqrt{\frac{166}{15}}.$$

Example 2.35 (A function not belonging to $W^{k,p}$ for $k > 1$). We again consider the function

$$u(x) = |x|, \quad x \in (-1, 1).$$

From Example 2.30, we know that its weak first derivative is

$$u'(x) = \operatorname{sgn}(x).$$

Since both u and u' belong to $L^p(-1, 1)$ for any $1 \leq p \leq \infty$, it follows that

$$u \in W^{1,p}(-1, 1).$$

However, to belong to $W^{2,p}(-1, 1)$, the function must also allow a second weak derivative belonging to $L^p(-1, 1)$. As demonstrated in Example 1.29 of Dacorogna [4, p. 30], differentiating u' gives

$$u'' = 2\delta_0,$$

where δ_0 denotes the Dirac mass. The Dirac mass is not an L^p therefore,

$$u \notin W^{2,p}(-1, 1).$$

we conclude that

$$u \notin W^{k,p}(-1, 1) \quad \text{for all } k > 1.$$

This example shows that increasing k asks for stronger regularity on a function. Even though $u(x) = |x|$ has a weak first derivative and belongs to $W^{1,p}(-1, 1)$, it does not allow a second weak derivative in $L^p(-1, 1)$ and so it cannot belong to any $W^{k,p}(-1, 1)$ with $k > 1$.

Another useful theorem which we will use is the Sobolev imbedding theorem, we only care for case 3 from Theorem 1.44 from Dacorogna [4].

Theorem 2.36 (Sobolev embedding theorem). *Let $\Omega \subset \mathbb{R}^2$ be a bounded open set with Lipschitz boundary.*

Case 3. If $p > n$ then

$$W^{1,p}(\Omega) \subset C^{0,\alpha}(\overline{\Omega}), \quad \text{for every } \alpha \in [0, 1 - n/p].$$

2.4.2 Poincaré Inequality

Theorem 2.37. *Let $\Omega \subset \mathbb{R}^n$ be a bounded open set and $1 \leq p \leq \infty$. Then there exists $\gamma = \gamma(\Omega, p) > 0$ so that,*

$$\|u\|_{L^p \text{ or } W^{1,p}} \leq \gamma \|\nabla u\|_{L^p} \quad \forall u \in W_0^{1,p}.$$

For a proof, see Dacorogna [4, p. 45].

2.5 Convex Analysis

Convex analysis is a broad area of mathematics, so this section focuses only on the concepts relevant to the Direct Method. We introduce the definition of convex set, convex

functions, and some conditions, alongside examples to illustrate these ideas.

2.5.1 Convex Sets

Definition 2.38. The set $\Omega \subset \mathbb{R}^n$ is said to be convex if for every $x, y \in \Omega$ and every $\lambda \in [0, 1]$ we have $\lambda x + (1 - \lambda)y \in \Omega$.

Example 2.39 (Convex Set Example). Let

$$\Omega = (0, 1).$$

Choose

$$x = 0.2, \quad y = 0.8, \quad \lambda = 0.25.$$

Then

$$\lambda x + (1 - \lambda)y = 0.25(0.2) + 0.75(0.8) = 0.65.$$

Since $0.65 \in (0, 1)$, the point remains in the set.

In fact, every weighted average of two points in $(0, 1)$ remains in $(0, 1)$. Therefore $(0, 1)$ is convex.

2.5.2 Convex Functions

Definition 2.40. Let $\Omega \subset \mathbb{R}^n$ be convex. The Function $f : \Omega \rightarrow \mathbb{R}$ is said to be convex if for every $x, y \in \Omega$ and every $\lambda \in [0, 1]$, the following inequality holds

$$f(x) \geq f(y) + \langle \nabla f(y); x - y \rangle, \forall x, y \in \mathbb{R}^n$$

Example 2.41. Let $\Omega = \mathbb{R}^n$ and define

$$f(x) = |x|^2.$$

Then

$$\nabla f(x) = 2x.$$

To verify that f is convex, take $x, y \in \mathbb{R}^n$. The convexity condition for a differentiable function is

$$f(x) \geq f(y) + \langle \nabla f(y), x - y \rangle.$$

Substituting $f(x) = |x|^2$ and $\nabla f(y) = 2y$ gives

$$|x|^2 \geq |y|^2 + 2\langle y, x - y \rangle.$$

Rearranging,

$$|x|^2 - |y|^2 - 2\langle y, x - y \rangle = |x - y|^2 \geq 0.$$

Hence the convexity inequality holds, and therefore

$$f(x) = |x|^2$$

is convex.

We can also verify this visually, as illustrated in Figure 2.2, it can be observed that the function $f(x) = |x|^2$ is convex.

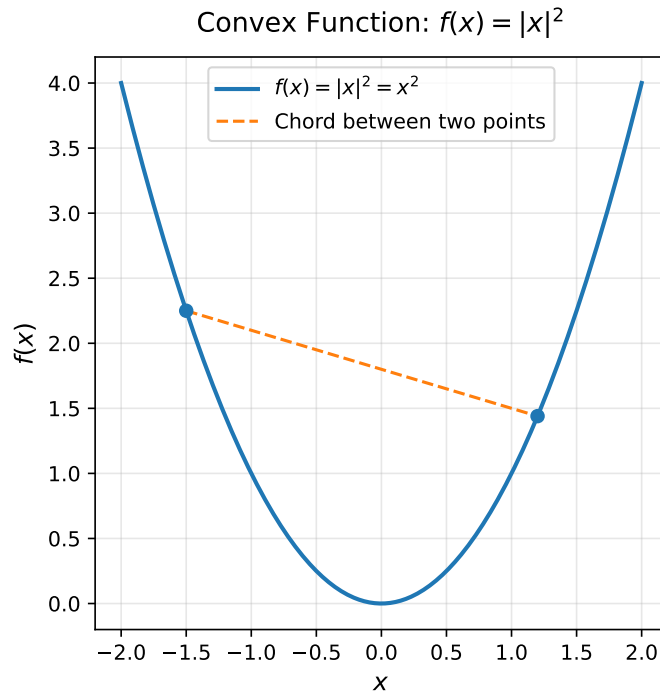


Figure 2.2: The function $f(x) = |x|^2$, with a chord illustrating its convexity.

2.5.3 Convexity Conditions

Theorem 2.42. *Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and $f \in C^1(\mathbb{R})$. We denote the scalar product in $\mathbb{R}^n$ by $\langle \cdot, \cdot \rangle$. The following assertions are equivalent.*

- (i) *the function f is convex;*

(ii) for every $x, y \in \mathbb{R}^n$, the following inequality holds

$$f(x) \geq f(y) + \langle \nabla f(y); x - y \rangle;$$

(iii) for every $x, y \in \mathbb{R}^n$, the following inequality is vails

$$\langle \nabla f(x) - \nabla f(y); x - y \rangle \geq 0;$$

(iv) If $f \in C^2(\mathbb{R}^n)$, then for every $x, v \in \mathbb{R}^n$,

$$\langle \nabla^2 f(x)v, v \rangle \geq 0.$$

This is equivalent to saying that the Hessian matrix $\nabla^2 f(x)$ is positive semidefinite.

2.6 The Direct Method

In this section, we develop the Direct Method of the Calculus of Variations. The Direct Method provides a framework for proving the existence of minimisers for a given functional and the conditions under which minimisers exist. We start by extending the model problem, (B), introduced at the beginning of the chapter, building on the theory of Sobolev spaces developed in Section 2.4. We then consider model problems to develop an intuition for the method. Following this, we introduce the main existence results and derive the Euler–Lagrange equation for the Direct Method in the setting of $\bar{u} \in W^{1,p}$. As with the previous sections, we will primarily follow Dacorogna [4], particularly Chapter 3, while also drawing on examples and notation from Rindler [7].

For the Direct Method, we consider the problem

$$\inf \left\{ I(u) = \int_{\Omega} f(x, u(x), \nabla u(x)) dx : u \in u_0 + W_0^{1,p}(\Omega) \right\} = m, \quad (\mathbf{P}_{\text{direct}})$$

where

- $\Omega \subset \mathbb{R}^n$ is a bounded open set;
- $f : \bar{\Omega} \times \mathbb{R} \times \mathbb{R}^n \rightarrow \mathbb{R}$, with $f = f(x, u, \xi)$;
- $u \in u_0 \in W_0^{1,p}(\Omega)$ means that $u, u_0 \in W_0^{1,p}(\Omega)$ and $u - u_0 \in W_0^{1,p}(\Omega)$. (this corresponds to $u = u_0$ on $\partial\Omega$.)

We will show that, under suitable hypotheses, the problem, P_{direct} , admits a solution $\bar{u} \in u_0 + W_0^{1,p}(\Omega)$. In particular, we will require the following two main hypotheses:

1. *Convexity*: for every $(x, u) \in \bar{\Omega} \times \mathbb{R}$, the function $\xi \mapsto f(x, u, \xi)$ is convex.
2. *Coercivity*: there exist $p > q \geq 1$ and constants $\alpha_1 > 0, \alpha_2, \alpha_3 \in \mathbb{R}$ such that

$$f(x, u, \xi) \geq \alpha_1 |\xi|^p + \alpha_2 |u|^q + \alpha_3,$$

for all $(x, u, \xi) \in \bar{\Omega} \times \mathbb{R} \times \mathbb{R}^n$.

The *Dirichlet integral* which has integrand

$$f(x, u, \xi) = \frac{1}{2} |\xi|^2$$

satisfies both hypotheses.

However the *minimal surface* problem whose integrand is given by

$$f(x, u, \xi) = \sqrt{1 + |\xi|^2}$$

satisfies Convexity but verifies Coercivity only with $p = 1$

2.6.1 The Model Case

We consider the Dirichlet integral,

$$\frac{1}{2} \int_{\Omega} |\nabla u(x)|^2 dx. \tag{2.3}$$

Theorem 2.43. *Let $\Omega \subset \mathbb{R}^n$ be bounded open set with Lipschitz boundary and $u_0 \in W^{1,2}(\Omega)$. The problem*

$$\inf \left\{ I(u) = \frac{1}{2} \int_{\Omega} |\nabla u(x)|^2 dx : u \in u_0 + W_0^{1,2}(\Omega) \right\} = m \tag{D}$$

has one and only one solution $\bar{u} \in u_0 + W_0^{1,2}(\Omega)$.

Furthermore $\bar{u}$ satisfies the weak form of the Laplace equation, namely

$$\int_{\Omega} \langle \nabla \bar{u}(x); \nabla \varphi(x) \rangle dx = 0, \forall \varphi \in W_0^{1,2}(\Omega) \tag{2.4}$$

where $\langle \cdot, \cdot \rangle$ denotes the scalar product in $\mathbb{R}^n$. Conversely, if $\bar{u} \in u_0 + W_0^{1,2}(\Omega)$ satisfies (2.4) then it is a minimiser of (D)

Remark 2.44. An interesting historical connection to Section 1.1 is that the existence theorem stated above, Theorem 2.43, was proved before the development of Sobolev spaces. Hilbert, Lebesgue, and Tonelli established existence results for variational problems using the mathematical framework available at the time, with the modern Sobolev-space formulation developing later. See [4].

Proof. **Part 1 (Existence).** We divide the proof into three steps.

(i) *Compactness*

Since $u_0 \in u_0 + W_0^{1,2}(\Omega)$ we have

$$0 \leq m \leq I(u_0) \leq \infty.$$

Let $u_\nu \in u_0 + W_0^{1,2}(\Omega)$ be a minimising sequence of (D), this means that

$$I(u_\nu) \rightarrow \inf\{I(u)\} = m, \text{ as } \nu \rightarrow \infty.$$

Observe that by the Poincaré inequality, Theorem 2.37, we can find constants $\gamma_1, \gamma_2 > 0$ so that,

$$\sqrt{2I(u_\nu)} = \|\nabla u_\nu\|_{L^2} \geq \gamma_1 \|u_\nu\|_{W^{1,2}} - \gamma_2 \|u_0\|_{W^{1,2}}.$$

Since u_ν is a minimising sequence and $m < \infty$ we deduced there exists $\gamma_3 > 0$ so that

$$\|u_\nu\| \leq \gamma_3.$$

It follows from Exercise 1.4.5 in Dacorogna [4, p. 47] that there exists, $\bar{u} \in u_0 + W_0^{1,2}(\Omega)$ and a subsequence u_ν so that

$$u_\nu \rightharpoonup \bar{u} \text{ in } W^{1,2}, \text{ as } v \rightarrow \infty.$$

(ii) *Lower Semicontinuity*

We will show that I is sequentially weakly lower semicontinuous; meaning,

$$u_\nu \rightharpoonup \bar{u} \text{ in } W^{1,2} \implies \liminf_{\nu \rightarrow \infty} I(u_\nu) \geq I(\bar{u}).$$

By expanding the gradient square, we get

$$\begin{aligned} |\nabla u_\nu|^2 &= |\nabla \bar{u}|^2 + 2\langle \nabla \bar{u}; \nabla u_\nu - \nabla \bar{u} \rangle + |\nabla u_\nu - \nabla \bar{u}|^2, \\ &\geq |\nabla \bar{u}|^2 + 2\langle \nabla \bar{u}; \nabla u_\nu - \nabla \bar{u} \rangle. \end{aligned}$$

Integrating this, we have

$$I(u_\nu) \geq I(\bar{u}) + \int_{\Omega} \langle \nabla \bar{u}; \nabla u_\nu - \nabla \bar{u} \rangle dx.$$

We need the second term to tend to 0. Since $\nabla \bar{u} \in L^2$ and $\nabla u_\nu - \nabla \bar{u} \rightharpoonup 0$ in L^2 by definition of weak convergence in L^2 , that

$$\lim_{\nu \rightarrow \infty} \int_{\Omega} \langle \nabla \bar{u}; \nabla u_\nu - \nabla \bar{u} \rangle dx = 0.$$

Therefore,

$$\liminf_{\nu \rightarrow \infty} I(u_\nu) \geq I(\bar{u}).$$

(iii) *Combine*

Since $\{u_\nu\}$ was a minimising sequence (i.e. $I(u_\nu) \rightarrow \inf\{I(u)\} = m$) and for such a sequence we have lower semicontinuity (i.e. $\liminf I(u_\nu) \geq I(\bar{u})$) we can deduce that $I(\bar{u}) = m$

Part 2 (Uniqueness).

Assume that there exist $\bar{u}, \bar{v} \in u_0 + W_0^{1,2}(\Omega)$ such that

$$I(\bar{u}) = I(\bar{v}) = m.$$

Since the function $\xi \mapsto |\xi|^2$ is strictly convex, the functional I is strictly convex. Therefore,

$$I\left(\frac{\bar{u} + \bar{v}}{2}\right) < \frac{1}{2}I(\bar{u}) + \frac{1}{2}I(\bar{v}) = m,$$

unless $\bar{u} = \bar{v}$. This contradicts the definition of m as the minimum of I . Hence,

$$\bar{u} = \bar{v}.$$

Therefore, the minimiser is unique.

Part 3 (Euler–Lagrange equation).

We want to establish (2.4). Let $\epsilon \in \mathbb{R}$ and $\varphi \in W_0^{1,2}(\Omega)$ be arbitrary. Since $\bar{u} \in$

$u_0 + W_0^{1,2}(\Omega)$ and $\varphi \in W_0^{1,2}(\Omega)$, we have

$$\bar{u} + \epsilon\varphi \in u_0 + W_0^{1,2}(\Omega).$$

Combined with $\bar{u}$ is minimiser of (D) leads to

$$\begin{aligned} I(\bar{u}) &\leq I(\bar{u} + \epsilon\varphi), \\ &= \int_{\Omega} \frac{1}{2} |\nabla \bar{u} + \epsilon \nabla \varphi|^2 dx, \\ &= I(\bar{u}) + \epsilon \int_{\Omega} \langle \nabla \bar{u}; \nabla \varphi \rangle dx + \epsilon^2 I(\varphi). \end{aligned}$$

As ϵ is arbitrary this leads to (2.4) which expresses,

$$\left. \frac{d}{d\epsilon} I(\bar{u} + \epsilon\varphi) \right|_{\epsilon=0} = 0.$$

□

2.6.2 A General Existence Theorem

In this section we state the direct methods theorem for $f \in C^0(\bar{\Omega} \times \mathbb{R} \times \mathbb{R}^n)$ and examples.

Theorem 2.45. *Let $\Omega \subset \mathbb{R}^n$ be a bounded open set with Lipschitz boundary. Let $f \in C^0(\bar{\Omega} \times \mathbb{R} \times \mathbb{R}^n)$, $f = f(x, u, \xi)$ satisfy,*

$$(H1) \quad \xi \mapsto f(x, u, \xi) \text{ is convex for every } (x, u) \in \bar{\Omega} \times \mathbb{R},$$

$$(H2) \quad \text{There exist } p > q \geq 1, \alpha_1 > 0, \alpha_2, \alpha_3 \in \mathbb{R} \text{ such that}$$

$$f(x, u, \xi) \geq \alpha_1 |\xi|^p + \alpha_2 |u|^q + \alpha_3, \quad \forall (x, u, \xi) \in \bar{\Omega} \times \mathbb{R} \times \mathbb{R}^n.$$

Let

$$(P) \quad \inf \left\{ I(u) = \int_{\Omega} f(x, u(x), \nabla u(x)) dx : u \in u_0 + W_0^{1,p}(\Omega) \right\} = m.$$

where $u_0 \in W^{1,p}(\Omega)$ with $I(u_0) < \infty$. Then there exists $\bar{u} \in u_0 + W_0^{1,p}(\Omega)$ a minimiser of (P). Furthermore if $(u, \xi) \rightarrow f(x, u, \xi)$ is strictly convex for every $x \in \bar{\Omega}$, then the minimiser is unique.

For a proof, see Dacorogna [4, p. 107].

Example 2.46. The Dirichlet integral satisfies all the hypotheses with $p = 2$. The generalisation of the Dirichlet integral is, where g is continuous and non negative and $p > 1$

$$f(x, u, \xi) = \frac{1}{p} |\xi|^p + g(x, u).$$

Example 2.47. The minimal surface problem has an integrand given by

$$f(x, u, \xi) = f(\xi) = \sqrt{1 + |\xi|^2}$$

satisfies all but $(H2)$, this is only verified at $p = 1$. But this creates the Sobolev space $W^{1,1}$ which is not reflexive.

Example 2.48. The weakening of $(H2)$ leads to the following counterexample.

Let $n = 1$,

$$f(x, u, \xi) = f(u, \xi) = \sqrt{u^2 + \xi^2},$$

over $(0, 1)$ with boundary $u(0) = 0, u(1) = 1$. with $u \in W^{1,1}(0, 1)$.

First, show that $m = 1$ we start by observing that $m \geq 1$,

$$I(u) \geq \int_0^1 |u'(x)| dx \geq \int_0^1 u'(x) dx = u(1) - u(0) = 1.$$

To establish that $m = 1$ we construct a minimising sequence u_ν .

$$u_\nu(x) = \begin{cases} 0, & \text{if } x \in [0, 1 - \frac{1}{\nu}] , \\ 1 + \nu(x - 1), & \text{if } x \in [1 - \frac{1}{\nu}, 1] . \end{cases}$$

Therefore $m = 1$ since

$$1 \leq I(u_\nu) = \int_{1-\frac{1}{\nu}}^1 \sqrt{(1 + \nu(x - 1))^2 + \nu^2} dx \leq \frac{1}{\nu} \sqrt{1 + \nu^2} \rightarrow 1, \text{ as } \nu \rightarrow \infty$$

Assume for the sake of contradiction there exists $\bar{u}$ a minimiser of (P) . We have

$$1 = I(\bar{u}) = \int_0^1 \sqrt{\bar{u}^2 + \bar{u}'^2} dx \geq \int_0^1 \bar{u} dx = \bar{u}(1) - \bar{u}(0) = 1$$

this implies that $\bar{u} = 0$ in $(0, 1)$. $\bar{u} \equiv 0$ is incompatible with boundary conditions. Thus (P) has no solution.

2.6.3 Euler–Lagrange Equation

We now extend the Euler–Lagrange Theorem 2.1 in the setting of the Direct Method, motivated by the general existence theorem, Theorem 2.45. This expansion allows for minimisers in the setting of $\bar{u} \in W^{1,p}(\Omega)$.

Theorem 2.49. *Let $\Omega \in \mathbb{R}^n$ be a bounded open set with Lipschitz boundary. Let $p \geq 1$ and $f \in C^1(\bar{\Omega} \times \mathbb{R} \times \mathbb{R}^n)$, $f = (x, u, \xi)$, satisfy*

$$(H3) \quad \text{There exists } b \geq 0 \text{ so that for every } (x, u, \xi) \in \bar{\Omega} \times \mathbb{R} \times \mathbb{R}^n$$

$$|f_u(x, u, \xi)|, |f_\xi(x, u, \xi)| \leq \beta(1 + |u|^{p-1} + |\xi|^{p-1})$$

where $f_\xi = (f_{\xi_1}, \dots, f_{\xi_n})$, $f_{\xi_i} = \partial f / \partial \xi_i$ and $f_u = \partial f / \partial u$.

Let $\bar{u} \in u_0 + W_0^{1,p}(\Omega)$ be a minimiser of

$$(P) \quad \inf \left\{ I(u) = \int_{\Omega} f(x, u(x), \nabla u(x)) dx : u \in u_0 + W_0^{1,p} \right\} = m$$

where $u_0 \in W^{1,p}(\Omega)$, the $\bar{u}$ satisfies the weak form of the Euler-Lagrange equation

$$(E_w) \quad \int_{\Omega} [f_u(x, \bar{u}, \nabla \bar{u})\varphi + \langle f_\xi(x, \bar{u}, \nabla \bar{u}); \nabla \varphi \rangle] dx = 0, \quad \forall \varphi \in W_0^{1,p}(\Omega).$$

Moreover, if $f \in C^2(\bar{\Omega} \times \mathbb{R} \times \mathbb{R}^n)$ and $\bar{u} \in C^2$ then $\bar{u}$ satisfies the standard Euler-Lagrange equation

$$(E) \quad \sum_{i=1}^n \frac{\partial}{\partial x_i} [f_{\xi_i}(x, \bar{u}, \nabla \bar{u})] = f_u(x, \bar{u}, \nabla \bar{u}), \quad \forall x \in \bar{\Omega}.$$

Conversely, if $(u, \xi) \rightarrow f(x, u, \xi)$ is convex for every $x \in \bar{\Omega}$ and if $\bar{u}$ is a solution of either (E_w) or (E) then it is a minimiser of (P)

For a proof, see Dacorogna [4, p. 113].

Chapter 3

Mathematical Analysis

In this chapter, we introduce the problem studied in this dissertation as an application of the Calculus of Variations and derive the corresponding mathematical model. We then apply the Direct Method of the Calculus of Variations to establish the existence of a minimiser and derive the associated Euler–Lagrange equation. Throughout this chapter, the problem is formulated using the arc-length parameter s , leading to the variational problem

$$J[\gamma(s)] = \min_{\gamma \in \Omega} \int_{s_0}^{s_1} (w_c c(\gamma(s)) + \lambda |\gamma''(s)|^2 + \mu) ds. \quad (\text{P})$$

3.1 Mathematical modelling

We consider a path-planning problem in a two-dimensional domain representing a terrain. The objective is to determine a path between two boundary points while balancing three competing considerations: the cost associated with the terrain traversed, the cost from the length of the path, and the cost of a curvature penalty along the path.

Let $\Omega \subset \mathbb{R}^2$ denote the domain representing the terrain, and let $a, b \in \Omega$ denote the prescribed starting and ending points, respectively. An admissible path connecting these points is represented by a sufficiently smooth curve

$$\gamma : [s_0, s_1] \rightarrow \Omega, \quad \gamma(s) = (x(s), y(s)),$$

satisfying the boundary conditions

$$\gamma(s_0) = a, \quad \gamma(s_1) = b.$$

Here, s denotes the arc-length parameter along the curve.

The collection of admissible paths satisfying these boundary conditions, together with any additional constraints imposed by the terrain, is denoted by Γ .

The terrain is characterised by a cost field

$$c : \Omega \rightarrow \mathbb{R}, \quad \Omega \subset \mathbb{R}^2.$$

where $c(x, y)$ represents the cost associated with traversing the terrain at the point (x, y) . Regions with larger values of $c(x, y)$ are therefore more costly for the path to traverse. The cost incurred by a path due solely to the terrain is given by the line integral

$$J_{\text{terrain}}(\gamma) = w_c \int_{\gamma} c(x, y), ds. \quad (3.1)$$

While minimising this quantity favours paths through regions of low terrain cost, it does not by itself account for the cost induced by length or curvature of the path. A path could, for example, reduce its terrain cost by taking a substantially longer route. Similarly, a path with large changes in direction may be undesirable even if it has low terrain cost and relatively short length as this is not realistic for a railway. We incorporate additional penalties for path length and curvature.

The resulting functional is defined by

$$J(\gamma) = \int_{\gamma} [w_c c(x, y) + \lambda \kappa^2 + \mu] ds, \quad (3.2)$$

where w_c , λ , and μ are non-negative parameters controlling the contribution of the terms of the integrand. These terms represent terrain cost, curvature, and length, respectively.

The path-planning problem is consequently formulated as the minimisation problem

$$\inf_{\gamma \in \Gamma} J(\gamma). \quad (3.3)$$

We now reformulate the integrand in (3.2) term by term to obtain the more convenient formulation given in (P).

3.1.1 Terrain Cost Term

The first term in the integrand of (3.2) represents the cost associated with traversing the terrain. Let

$$\gamma(s) = (x(s), y(s)),$$

denote the path parametrised by arc length s . Since the cost field is defined by

$$c : \Omega \rightarrow \mathbb{R},$$

the cost at any point on the path is obtained by evaluating c at the position of the curve. Thus,

$$c(x, y) \longrightarrow c(x(s), y(s)) = c(\gamma(s)).$$

The key step here is the composition of the cost field with the path,

$$c \circ \gamma : [s_0, s_1] \rightarrow \mathbb{R}, \quad (c \circ \gamma)(s) = c(\gamma(s)).$$

In other words, the terrain cost along the path is represented by the composition $c \circ \gamma$. This makes explicit why the cost field $c(x, y)$ becomes $c(\gamma(s))$ when the path is parametrised by s . rather than evaluating the cost at an arbitrary point in the domain, we evaluate it at the position of the curve as it traverses the terrain.

Consequently, the terrain component of the functional becomes

$$w_c \int_{\gamma} c(x, y) ds = w_c \int_{s_0}^{s_1} c(\gamma(s)) ds.$$

The parameter $w_c \geq 0$ controls the relative importance of the cost field within the functional.

3.1.2 Path Length Term

The final term in the integrand of (3.2) represents a penalty on the total length of the path. For a curve

$$\gamma : [s_0, s_1] \rightarrow \mathbb{R}^2,$$

the length of the path is given by

$$L(\gamma) = \int_{s_1}^{s_0} |\gamma'(s)| ds, \tag{3.4}$$

where s denotes the arc-length parameter.

More generally, if the curve is initially parametrised by an arbitrary parameter $t \in [t_0, t_1]$, its length is given by

$$L(\gamma) = \int_{t_0}^{t_1} |\gamma'(t)| dt.$$

Introducing the arc-length parameter

$$s(t) = \int_{t_0}^t |\gamma'(\tau)| d\tau,$$

allows the curve to be parametrised directly by the distance travelled along the path. Consequently,

$$\left| \frac{d\gamma}{ds} \right| = 1.$$

Thus, for an arc-length parametrised curve, an change in ds corresponds directly to an change in the length of the path, giving

$$L(\gamma) = \int_{s_0}^{s_1} 1 ds.$$

The length penalty in the functional is therefore

$$J_{\text{length}}(\gamma) = \mu L(\gamma) = \mu \int_{s_0}^{s_1} 1, ds. \quad (3.5)$$

Hence, the contribution of the length penalty to the integrand of (3.2) is simply

$$\mu.$$

The parameter $\mu \geq 0$ controls the relative importance assigned to the total length of the path. Increasing μ therefore places a greater penalty on longer paths.

3.1.3 Curvature Cost Term

The second term in the integrand of (3.2) penalises the curvature of the path. To express this term in terms of the derivatives of γ , we first consider a general parametrisation

$$\gamma(t) = (x(t), y(t)), \quad t \in [t_0, t_1].$$

For a sufficiently smooth planar curve, the curvature is given by

$$\kappa(t) = \frac{|\gamma'(t) \times \gamma''(t)|}{|\gamma'(t)|^3}. \quad (3.6)$$

We now consider the arc-length parametrisation used in the formulation of the functional. As shown in the previous subsection, when s is the arc-length parameter, the curve

has unit speed, so that

$$|\gamma'(s)| = 1. \quad (3.7)$$

It can be shown that, $\gamma'(s)$ and $\gamma''(s)$ are orthogonal. Consequently,

$$|\gamma'(s) \times \gamma''(s)| = |\gamma'(s)| |\gamma''(s)| = |\gamma''(s)|,$$

where (3.7) has been used. Substituting these results into (3.6) gives

$$\kappa(s) = \frac{|\gamma''(s)|}{|\gamma'(s)|^3} = \frac{|\gamma''(s)|}{(1)^3} = |\gamma''(s)|.$$

Therefore,

$$\kappa(s) = |\gamma''(s)|. \quad (3.8)$$

The curvature contribution to the functional is consequently

$$J_{\text{curvature}}(\gamma) = \lambda \int_{\gamma} \kappa^2 ds. \quad (3.9)$$

Using (3.8), this becomes

$$J_{\text{curvature}}(\gamma) = \lambda \int_{s_0}^{s_1} |\gamma''(s)|^2 ds.$$

Hence, the curvature contribution to the integrand of (3.2) can be written as

$$\lambda \kappa^2 = \lambda |\gamma''(s)|^2.$$

The use of κ^2 penalizes curvature independently of its direction, while assigning a greater cost to regions where the path bends more sharply. The parameter $\lambda \geq 0$ therefore controls the relative importance of path smoothness within the functional.

3.2 Euler–Lagrange Equation - Classical Approach

As established in the previous section, the proposed path-planning problem can be expressed in terms of the arc-length parameter s . Recalling (P), the functional is given by

$$J[\gamma] = \int_{s_0}^{s_1} (w_c c(\gamma(s)) + \lambda |\gamma''(s)|^2 + \mu) ds. \quad (3.10)$$

The corresponding Euler–Lagrange equation provides a necessary condition for a suf-

ficiently smooth minimiser of the functional. For a second-order functional of the form

$$J[\gamma] = \int_{s_0}^{s_1} F(\gamma, \gamma', \gamma'') ds,$$

the second-order Euler–Lagrange equation is given by

$$\frac{\partial F}{\partial \gamma} - \frac{d}{ds} \left(\frac{\partial F}{\partial \gamma'} \right) + \frac{d^2}{ds^2} \left(\frac{\partial F}{\partial \gamma''} \right) = 0. \quad (3.11)$$

For the functional $F(\gamma, \gamma', \gamma'') = c(\gamma) + \lambda |\gamma''|^2 + \mu$, we obtain

$$\frac{\partial F}{\partial \gamma} = w_c \nabla c(\gamma), \quad \frac{\partial F}{\partial \gamma'} = 0, \quad \frac{\partial F}{\partial \gamma''} = 2\lambda \gamma''.$$

Substitution into the Euler–Lagrange equation therefore gives

$$w_c \nabla c(\gamma) + 2\lambda \gamma^{(4)} = 0,$$

or equivalently

$$\gamma^{(4)}(s) = -\frac{w_c}{2\lambda} \nabla c(\gamma(s)). \quad (3.12)$$

For convenience, define the constant

$$\alpha := \frac{w_c}{2\lambda}.$$

Thus, (3.12) can be written more compactly as

$$\gamma^{(4)}(s) = -\alpha \nabla c(\gamma(s)). \quad (3.13)$$

Equation (3.13) characterises the classical stationary paths associated with the proposed functional. In general, however, solving this equation explicitly is not straightforward. In particular, this equation is a fourth-order nonlinear system of ordinary differential equations..

3.2.1 Gaussian Hill

To illustrate the resulting nonlinearity, consider a uniform background cost containing a region of increased cost. A Gaussian hill provides a simple model for such a region, given by

$$c(\gamma(s)) = H \exp \left(-\frac{|\gamma(s) - p|^2}{2\sigma^2} \right), \quad (3.14)$$

where $H > 0$ represents the magnitude of the increase in cost, $p \in \mathbb{R}^2$ is the centre of the hill and $\sigma > 0$ controls its spatial spread. If the background cost is c_0 and the maximum cost is $c_{\max}$, then

$$H = c_{\max} - c_0.$$

The gradient of (3.14) with respect to γ is

$$\nabla c(\gamma) = -\frac{H}{\sigma^2} \exp\left(-\frac{|\gamma - p|^2}{2\sigma^2}\right) (\gamma - p).$$

Therefore, equation (3.13) becomes

$$\gamma^{(4)}(s) = \frac{\alpha H}{\sigma^2} \exp\left(-\frac{|\gamma(s) - p|^2}{2\sigma^2}\right) (\gamma(s) - p). \quad (3.15)$$

Even for this relatively simple cost landscape, the equation (3.15) is a nonlinear fourth-order system. Consequently, obtaining an explicit analytical solution is generally difficult and would require solving a nonlinear boundary-value problem. Rather than attempting to obtain a closed-form solution, the remainder of this chapter therefore investigates the problem from a variational perspective. In particular, we shall use the Direct Method of the Calculus of Variations to establish conditions which the functional admits a minimiser, without requiring that minimiser to be explicitly constructed. The Gaussian hill defined in (3.14) will also be used as the cost function in the numerical simulations presented later in this dissertation, (4.1). This provides a common cost landscape through which the analytical and numerical aspects of the problem can be considered.

3.2.2 Rational Quadratic

Considering how the Gaussian Hill, described in (3.14), leads to a difficult fourth-order ODE, we try a simpler function for c to check the Euler-ladgrange equation. We choose c ,

$$c(\gamma(s)) = c_0 + \frac{HR^2}{R^2 + |\gamma(s) - p|^2}, \quad (3.16)$$

Where $H > 0$ represents the magnitude of the increase in cost, $p \in \mathbb{R}^2$ is the centre of the hill and R is the radius. The background cost is c_0 and the maximum cost is $c_{\max}$, then

$$H = c_{\max} - c_0.$$

The gradient of (3.16) with respect to γ is

$$\nabla c(\gamma) = -\frac{2HR^2(\gamma - p)}{(R^2 + |\gamma - p|^2)^2}.$$

Therefore, the Euler–Lagrange equation, (3.13), becomes

$$\gamma^{(4)}(s) = \frac{(2HR^2\alpha)(\gamma(s) - p)}{(R^2 + |\gamma(s) - p|^2)^2}. \quad (3.17)$$

This is a non-linear fourth-order ordinary differential equation. Although the order of the equation cannot be reduced directly from the form of the right-hand side, its non-linear behaviour can be simplified asymptotically. In particular, when

$$|\gamma(s) - p|^2 \gg R^2,$$

the denominator satisfies

$$(R^2 + |\gamma(s) - p|^2)^2 \sim |\gamma(s) - p|^4.$$

Consequently,

$$\gamma^{(4)}(s) \sim \frac{(2HR^2\alpha)(\gamma(s) - p)}{|\gamma(s) - p|^4}.$$

In the scalar case, this has the asymptotic behaviour

$$\gamma^{(4)}(s) \sim \frac{C}{(\gamma(s) - p)^3} \sim \frac{1}{\gamma^3},$$

for an appropriate constant C . Thus, far from the point p , the forcing term decays cubically with the distance from p . This provides a much simpler ODE to solve and analyse.

Although (3.17) is a non-linear fourth-order ordinary differential equation, its right-hand side is a smooth function of γ . Consequently, the equation may be revised as a system of first-order ordinary differential equations, to which the Picard–Lindelöf theorem, Theorem 2.2 [9], applies. It follows that, for prescribed initial data

$$\gamma(s_0), \quad \gamma'(s_0), \quad \gamma''(s_0), \quad \gamma'''(s_0),$$

there exists a unique local solution to (3.17). Thus, despite the non-linear nature of the Euler–Lagrange equation, the corresponding initial-value problem admits a unique local solution.

The quadratic rational hill was also tested numerically, producing results comparable to those obtained for the Gaussian hill. A detailed discussion of Gaussian hill experiments is deferred to Chapter 4. I will not be describing the quadratic rational hill in details however the implementation and numerical experiments are provided in the accompanying GitHub repository [8].

3.3 Existence Theorem - Direct Approach

In this chapter, we set out an existence theorem for our path planning problem, together with its proof. The theorem focuses on the rational hill function for c given by (3.19). We begin with a brief discussion of the challenges involved in establishing existence for this problem and why the general theorems presented by Dacorogna cannot be applied directly. We then formulate the theorem, specifying the different components of the functional and the assumptions required. The main idea behind the theorem is that a minimiser of the functional exists within the chosen admissible set. Finally, we prove the result by following a similar approach to that used in the proof of existence for the Dirichlet integral in Theorem 2.43.

Remark 3.1. Before we state and prove the existence theorem for problem (P) we note that it is not possible to apply the general existence theorem for Calculus of Variations given in Section 2.6 This is because the integrand in (P) contains higher order derivative term due to the curvature. As a result the minimization problem differs fundamentally from the minimization problem in Theorem 2.45. There are numerous studies in the literature on minimization problems with functionals including curvature. However, their setting and the analysis involved is out of the scope of this dissertation. For completeness, we refer the interested readers to the work of Boscain, Charlot and Rossi [10], and Bellettini [3].

We now state and prove the existence result.

Theorem 3.2. *We consider the functional*

$$J[\gamma] = \int_{s_0}^{s_1} (w_c c(\gamma(s)) + \lambda |\gamma''(s)|^2 + \mu) ds, \quad (3.18)$$

where $s \in [s_0, s_1]$ denotes the curve parameter and $\gamma : [s_0, s_1] \rightarrow \mathbb{R}^2$ represents the parametrised path. The terrain cost is modelled by a rational hill of the form

$$c(x) = c_0 + \frac{HR^2}{R^2 + |x - p|^2}, \quad p \in \mathbb{R}^2, \quad (3.19)$$

where $c_0 \geq 0$ represents the baseline terrain cost, $H > 0$ represents the maximum additional cost associated with the hill, $R > 0$ determines its characteristic radius, and p denotes the centre of the hill.

Let $s_0 < s_1$, let $A, B \in \mathbb{R}^2$ be prescribed boundary points, and let $v_0, v_1 \in \mathbb{R}^2$ be prescribed boundary velocities. We define the admissible class

$$\mathcal{A} = \{ \gamma \in W^{2,2}((s_0, s_1); \mathbb{R}^2) : \gamma(s_0) = A, \gamma(s_1) = B, \gamma'(s_0) = v_0, \gamma'(s_1) = v_1 \}. \quad (3.20)$$

We define the constants so that

$$w_c \geq 0, \quad \lambda > 0, \quad \mu \geq 0.$$

We assume $\mathcal{A} \neq \emptyset$. Then the minimisation problem

$$\inf_{\gamma \in \mathcal{A}} J[\gamma] \quad (3.21)$$

admits a minimiser $\bar{\gamma} \in \mathcal{A}$.

Proof. We divide the proof into three steps

(i) *Compactness*

Let

$$m = \inf_{\gamma \in \mathcal{A}} J[\gamma].$$

Since $\mathcal{A} \neq \emptyset$, there exists $\gamma_0 \in \mathcal{A}$ and hence

$$m \leq J[\gamma_0] < \infty.$$

Since $H > 0$ and $R > 0$,

$$c(\gamma) = c_0 + \frac{HR^2}{R^2 + |\gamma - p|^2} \geq c_0.$$

Therefore,

$$J[\gamma] \geq \lambda \int_{s_0}^{s_1} |\gamma''(s)|^2 ds + (w_c c_0 + \mu)(s_1 - s_0).$$

We let $\gamma_\nu \subset \mathcal{A}$ be a minimising sequence, so that

$$J[\gamma_\nu] \longrightarrow m \quad \text{as } \nu \rightarrow \infty.$$

At some point we reach

$$J[\gamma_\nu] \leq m + 1.$$

Therefore,

$$\lambda \int_{s_0}^{s_1} |\gamma''(s)|^2 ds \leq m + 1 - (w_c c_0 + \mu)(s_1 - s_0).$$

Using the definition of the Lebesgue norm, Definition 2.18, choosing an appropriate constant C , alongside $\lambda > 0$ we have

$$\|\gamma''_\nu\|_{L^2} \leq \sqrt{C},$$

meaning $\|\gamma''_\nu\|_{L^2}$ is bounded. Since the endpoint values and endpoint derivatives are fixed, we can apply the Poincaré inequality, Theorem 2.37, subsequently on γ and γ' . This gives constants $C_1, C_2 > 0$ such that

$$\|\gamma_\nu\|_{W^{2,2}} \leq C_1 \|\gamma''_\nu\|_{L^2} + C_2.$$

Consequently, γ_ν is bounded in $W^{2,2}$.

Since $W^{2,2}((s_0, s_1); \mathbb{R}^2)$ is reflexive, there exists a subsequence, not relabelled, and some $\bar{\gamma} \in W^{2,2}((s_0, s_1); \mathbb{R}^2)$ such that

$$\gamma_\nu \rightharpoonup \bar{\gamma} \quad \text{weakly in } W^{2,2}.$$

Moreover, since

$$\gamma \in W^{2,2}((s_0, s_1); \mathbb{R}^2),$$

we also have

$$\gamma' \in W^{1,2}((s_0, s_1); \mathbb{R}^2).$$

Here, the domain of γ' is the one-dimensional interval $(s_0, s_1) \subset \mathbb{R}$, so $n = 1$, while the exponent in $W^{1,2}$ gives $p = 2$. Thus, applying Case 3 of the Sobolev embedding theorem, Theorem 2.36,

$$\gamma' \in C^{0,\alpha}([s_0, s_1]; \mathbb{R}^2), \quad 0 \leq \alpha \leq \frac{1}{2}.$$

Since $C^{0,\alpha} \subset C^0$ and that γ' is continuous, we have

$$\gamma \in C^1([s_0, s_1]; \mathbb{R}^2).$$

Therefore, the corresponding Sobolev embedding is compact, so

$$W^{2,2}((s_0, s_1); \mathbb{R}^2) \subset C^1([s_0, s_1]; \mathbb{R}^2).$$

After passing to a further subsequence if necessary,

$$\gamma_\nu \rightarrow \bar{\gamma} \quad \text{strongly in } C^1([s_0, s_1]; \mathbb{R}^2).$$

Due to the strong convergence in C^1 , the boundary conditions in $\mathcal{A}$ are preserved, and therefore

$$\bar{\gamma} \in \mathcal{A}.$$

(ii) *Lower Semicontinuity*

We want to show that

$$J[\bar{\gamma}] \leq \liminf_{\nu \rightarrow \infty} J[\gamma_\nu].$$

First consider the curvature term. Since

$$\gamma_\nu'' \rightharpoonup \bar{\gamma}'' \quad \text{weakly in } L^2,$$

and the function

$$x \mapsto |x|^2$$

is convex and weakly lower semicontinuous, we have

$$\int_{s_0}^{s_1} |\bar{\gamma}''(s)|^2 ds \leq \liminf_{\nu \rightarrow \infty} \int_{s_0}^{s_1} |\gamma_\nu''(s)|^2 ds.$$

Next consider the cost term. Since

$$\gamma_\nu \rightarrow \bar{\gamma} \quad \text{uniformly on } [s_0, s_1],$$

and c is continuous, we have

$$c(\gamma_\nu(s)) \rightarrow c(\bar{\gamma}(s))$$

uniformly on $[s_0, s_1]$. Consequently,

$$\int_{s_0}^{s_1} c(\gamma_\nu(s)) ds \longrightarrow \int_{s_0}^{s_1} c(\bar{\gamma}(s)) ds.$$

Combining these two results with the fact that for the final term we have,

$$\int_{s_0}^{s_1} \mu ds = \mu(s_1 - s_0).$$

We obtain

$$J[\bar{\gamma}] \leq \liminf_{\nu \rightarrow \infty} J[\gamma_\nu].$$

(iii) *Combine*

Since γ_ν is a minimising sequence,

$$J[\gamma_\nu] \longrightarrow m.$$

By weak lower semicontinuity,

$$J[\bar{\gamma}] \leq \liminf_{\nu \rightarrow \infty} J[\gamma_\nu] = m.$$

However, since $\bar{\gamma} \in \mathcal{A}$ and m is the infimum of J over $\mathcal{A}$,

$$J[\bar{\gamma}] \geq m.$$

Therefore,

$$J[\bar{\gamma}] = m.$$

Hence $\bar{\gamma}$ is a minimiser of (3.18).

□

From this proof, we have established that a minimiser exists for this path planning problem under the conditions set out. However, we do not know if this minimiser is unique or whether it satisfies the Euler–Lagrange equation. This could be investigated further. Nevertheless, knowing that a minimiser exists provides a theoretical basis for the numerical simulations considered in the following chapter. We can use this confidence to obtain an approximation to a minimising path.

Chapter 4

Numerical Methods

4.1 Introduction

In this chapter, we describe and evaluate the results of a numerical investigation into the optimisation of paths over a cost field. The numerical model aims to balance three competing objectives: the cost of traversing the terrain, the total length of the path, and its curvature. As discussed in the previous chapter, a useful interpretation of this problem is the placement of a railway track, where it is desirable to minimise the cost of the terrain through which the track passes while also keeping the total length and curvature of the track within reasonable bounds.

We consider the same model problem to which the Direct Method was applied in the previous chapter, given by (3.2).

We then describe the numerical method used to obtain approximate minimisers of the functional. The optimisation is performed using simulated annealing, with a segmented shift move used to generate neighbouring paths. This stochastic approach allows the algorithm to explore the space of admissible paths while providing the possibility of escaping local minima. We then describe the discretisation of the model problem and its implementation in Python, before presenting and analysing the numerical results obtained.

The complete source code and computational results used in this chapter are publicly available in the accompanying GitHub repository [8].

4.2 Numerical Discretisation

In order to investigate the model problem computationally, the continuous variational problem must first be converted into a discrete form which can be evaluated numerically.

In particular, both the terrain and the admissible paths must be represented using a finite number of discrete points. This provides a finite-dimensional approximation to the original optimisation problem, which can then be investigated using the simulated annealing method described in the following sections.

The numerical implementation was developed using Python within Jupyter notebooks. A class-based architecture was adopted to separate the representation of the terrain, paths, and associated cost calculations. The implementation was also modularised so that individual components of the numerical model could be modified and tested independently. The complete implementation is available in the accompanying GitHub repository [8].

4.2.1 Discretisation of the Terrain

The continuous terrain domain Ω is discretised into an $n \times n$ grid, with each grid point assigned a corresponding terrain cost $c(x, y)$. For the simulations considered in this chapter, the domain is taken to be

$$\Omega = [0, 100] \times [0, 100],$$

with a resolution of 100×100 grid points. Initially, a uniform background cost of $c_0 = 0.1$ is assigned to the entire domain.

To introduce variation in the terrain cost, a Gaussian-shaped hill is added to the cost field. The hill is defined by its centre (x_c, y_c) , radius r , and peak cost $c_{\max}$. Let

$$d(x, y) = \sqrt{(x - x_c)^2 + (y - y_c)^2},$$

denote the distance from (x, y) to the centre of the hill. The terrain cost within the specified radius is then given by

$$c(x, y) = c_0 + (c_{\max} - c_0) \exp\left(-\frac{d(x, y)^2}{2\sigma^2}\right), \quad (4.1)$$

where

$$\sigma = \frac{r}{3}.$$

The Gaussian profile produces the maximum cost at the centre of the hill, with the cost decreasing smoothly towards the edge of its specified radius. Outside this radius, the hill has no effect on the background terrain cost. For the terrain shown in Figure 4.1, the parameters used are

$$(x_c, y_c) = (50, 50), \quad r = 30, \quad c_{\max} = 0.8.$$

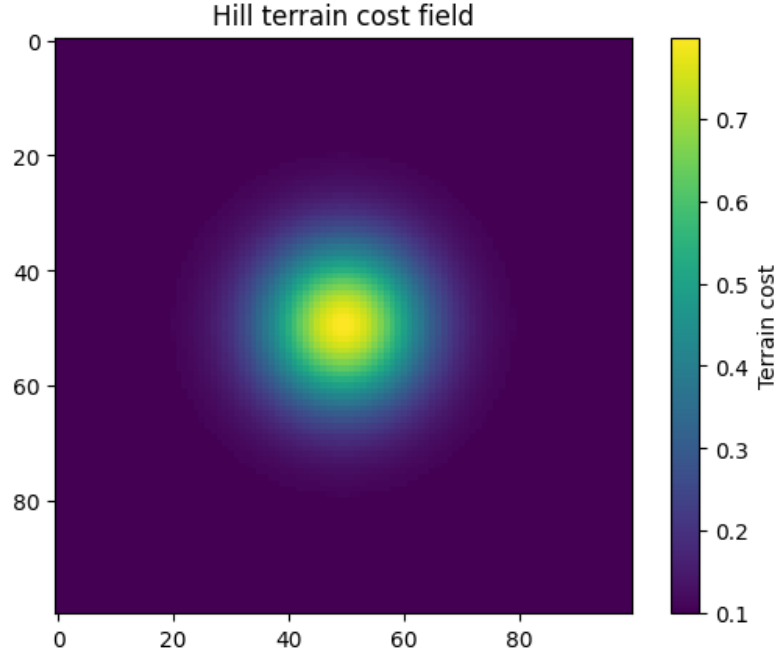


Figure 4.1: Discretised terrain cost field with a Gaussian-shaped hill.

4.2.2 Discretisation of the Path

The continuous curve γ is approximated by a finite sequence of ordered points

$$\gamma_h = \{p_0, p_1, \dots, p_N\}, \quad p_i = (x_i, y_i).$$

The endpoints are fixed at the prescribed boundary points,

$$p_0 = a, \quad p_N = b,$$

while the intermediate points are varied during optimisation. Consecutive points are joined by straight-line segments.

4.2.3 Path Constraints

Following the path discretisation, candidate paths are required to satisfy the constraints of the numerical model. Two conditions are imposed: the path must progress monotonically in the x -direction and must not enter any impassable region of the terrain.

The first condition requires

$$x_{i+1} > x_i, \quad i = 0, \dots, N-1, \quad (4.2)$$

which ensures that the path progresses from the start point towards the end point without reversing direction in the x -coordinate.

The second condition requires the path to remain outside any impassable region. Since checking only the discrete path points would not guarantee that the connecting line segments avoid an obstacle, the segments between consecutive points are sampled at a number of intermediate locations. A candidate path is considered feasible only if none of these sampled points lies within an impassable region.

These conditions are checked before a candidate path can be accepted by the simulated annealing algorithm. They therefore restrict the search space to paths which satisfy the required geometric and terrain constraints.

4.2.4 Cost Functional

The cost functional provides the objective used to evaluate each candidate path during the simulated annealing optimisation. For a discrete path γ_h , the three components of the continuous functional are calculated separately and then combined to give the total cost.

The terrain contribution is approximated by considering each segment between consecutive points. The terrain cost is evaluated at the midpoint of each segment and multiplied by its length, giving

$$C_{\text{terrain}}(\gamma_h) = \sum_{i=0}^{N-1} c\left(\frac{p_i + p_{i+1}}{2}\right) \|p_{i+1} - p_i\|. \quad (4.3)$$

This provides a numerical approximation to the terrain term

$$\int_{\gamma} c(x, y) ds.$$

The length of the discrete path is calculated as

$$C_{\text{length}}(\gamma_h) = \sum_{i=0}^{N-1} \|p_{i+1} - p_i\|, \quad (4.4)$$

which provides an approximation to the continuous path length $L(\gamma)$.

Finally, the curvature contribution is estimated using the second finite difference of

three consecutive path points:

$$q_i = p_{i+1} - 2p_i + p_{i-1}. \quad (4.5)$$

The magnitude of q_i measures the deviation of p_i from the straight line suggested by its neighbours. The discrete curvature cost is therefore calculated as

$$C_{\text{curvature}}(\gamma_h) = \sum_{i=1}^{N-1} \|p_{i+1} - 2p_i + p_{i-1}\|^2. \quad (4.6)$$

Straight sections of the path give a zero contribution, while sharper changes in direction produce a larger penalty. This provides a simple substitution for the continuous curvature term $\int_{\gamma} \kappa^2 ds$, although it does not explicitly account for the arc-length spacing between consecutive points.

The three components are then combined to form the discrete objective function

$$J_h(\gamma_h) = w_c C_{\text{terrain}}(\gamma_h) + \lambda C_{\text{curvature}}(\gamma_h) + \mu C_{\text{length}}(\gamma_h). \quad (4.7)$$

This is the quantity evaluated by the simulated annealing algorithm when comparing candidate paths.

4.3 Simulated Annealing

In this section, we describe the simulated annealing method used to optimise the discretised model problem, together with the choices made in its implementation and the resulting behaviour. The approach is inspired by the method presented by Amberg [1]. In their work, simulated annealing is applied to the capacitated minimum spanning tree problem. Having studied simulated annealing as part of the MSc module Stochastic Search and Optimisation, this provided motivation for applying the method to the path optimisation problem considered here.

4.3.1 Motivation

The stochastic approach used in this work is motivated by the distinction between proving the existence of a minimiser and actually finding one. The Direct Method of the Calculus of Variations can be used to establish the existence of minimisers for suitable functionals, but obtaining a minimiser explicitly can be considerably more challenging. For the model

problem (3.2), an analytical solution is generally difficult to obtain, particularly when the cost field $c(x, y)$ is complicated or not known in an explicit form.

Simulated annealing provides a practical approach to this problem by combining random exploration of the search space with a mechanism for gradually favouring lower-cost solutions. The method was also computationally suitable for the resources available for this project, allowing repeated simulations to be performed on a personal M4 MacBook Air.

A shift move was used to generate neighbouring paths. This provided a simple way of modifying individual points of the discretised path while also allowing the behaviour of the optimisation to be visualised during the simulation. This visualisation proved particularly useful for investigating the choice of the weighting parameters (w_c, λ, μ) and for identifying issues with the numerical implementation.

4.3.2 The Simulated Annealing Algorithm

Simulated annealing is an iterative stochastic optimisation method in which a sequence of candidate solutions is generated and evaluated. Starting from an initial feasible path, a neighbouring path is generated using the segmented shift move described in Section 4.4.3. The cost of the candidate is then compared with that of the current path.

If the candidate has a lower cost, it is accepted. A candidate with a higher cost may also be accepted according to the acceptance criterion described below. After each iteration, the temperature is reduced according to the chosen cooling schedule. Throughout the optimisation, the lowest-cost path encountered is stored as the best solution.

The main steps of the algorithm can therefore be summarised as:

1. Initialise the algorithm with a feasible path.
2. Generate a neighbouring candidate path.
3. Calculate the cost of the candidate using the discrete objective functional.
4. Check that the candidate satisfies the required path constraints.
5. Accept or reject the candidate according to the acceptance criterion.
6. Update the best path found so far.
7. Reduce the temperature and repeat until the specified number of iterations has been reached.

4.3.3 Acceptance Criterion

Let J_{current} denote the cost of the current path and $J_{\text{candidate}}$ the cost of a proposed path. The change in cost is defined as

$$\Delta J = J_{\text{candidate}} - J_{\text{current}}. \quad (4.8)$$

If $\Delta J < 0$, the candidate has a lower cost and is accepted. If $\Delta J \geq 0$, the candidate may still be accepted with probability

$$P(\text{accept}) = \exp\left(-\frac{\Delta J}{T}\right), \quad (4.9)$$

where T denotes the current temperature.

This allows the algorithm to occasionally accept higher-cost paths, particularly when the temperature is high. Such moves allow the search to move away from local minima and explore alternative regions of the solution space. As the temperature decreases, the probability of accepting higher-cost solutions also decreases, causing the algorithm to increasingly favour improvements.

In addition to this cost-based acceptance criterion, candidate paths must satisfy the validity conditions imposed by the numerical model. In particular, the path ordering and terrain constraints are checked before a candidate can be accepted. These constraints are considered in more detail in Section 4.2.3.

4.3.4 Temperature and Cooling Schedule

The temperature controls the degree of exploration performed by the simulated annealing algorithm. The initial temperature determines the likelihood of accepting higher-cost candidate paths at the beginning of the optimisation, while progressively reducing the temperature causes the algorithm to become increasingly selective.

In the implementation, the temperature is updated after each iteration according to the geometric cooling schedule

$$T_{k+1} = rT_k, \quad (4.10)$$

where r is the cooling rate. Starting from an initial temperature T_0 , this gives

$$T_k = T_0 r^k. \quad (4.11)$$

For the simulations presented in this chapter, the simulated annealing algorithm was

run for a specified number of iterations, with the initial temperature and cooling rate selected experimentally.

4.4 Neighbourhood Construction

At each iteration of the simulated annealing algorithm, a neighbouring path must be generated from the current path. The neighbourhood construction determines how the search explores the space of possible paths. Two approaches were considered: a basic shift move and a segmented shift move.

4.4.1 The Basic Shift Move

The initial neighbourhood construction used a shift move, in which a single interior waypoint of the path is selected and randomly perturbed. The displacement is sampled from a two-dimensional Gaussian distribution,

$$(\Delta x, \Delta y) \sim \mathcal{N}(0, \sigma^2 I).$$

The start and end points remain fixed, preserving the boundary conditions of the problem.

The waypoint is selected probabilistically, with points in regions of higher terrain cost given greater probability of being selected. The resulting candidate path is then evaluated using the discrete cost functional before being passed to the acceptance criterion of the simulated annealing algorithm.

4.4.2 Limitations of the Basic Shift Move

Although the basic shift move provides a simple method for exploring the space of paths, modifying only one waypoint at a time can make larger changes to the overall shape of the path difficult to achieve. In particular, substantial changes to a path may require many successive waypoint movements.

This becomes more significant when the path needs to move around a larger region of high-cost or impassable terrain. Moving individual points can produce inefficient intermediate configurations and can limit the rate at which the algorithm explores substantially different path geometries. These observations motivated the introduction of a move which could modify several neighbouring points simultaneously.

4.4.3 The Segmented Shift Move

The segmented shift move extends the basic shift move by perturbing a contiguous section of the path rather than a single waypoint. A segment of a prescribed length is selected from the interior points of the path. The starting position of the segment is selected probabilistically according to the average terrain cost of the points within the segment, again favouring regions of higher terrain cost.

Once selected, the entire segment is displaced by the same two-dimensional Gaussian random vector,

$$(\Delta x, \Delta y) \sim \mathcal{N}(0, \sigma^2 I).$$

The endpoints of the path remain fixed. The resulting candidate is then checked against the path constraints before its cost is evaluated and the simulated annealing acceptance criterion is applied.

Moving several neighbouring points together allows larger-scale changes in the geometry of the path while maintaining the local structure of the displaced section. This provides the algorithm with an additional mechanism for exploring substantially different path configurations compared with the basic shift move.

4.5 Numerical Results and Discussion

This section presents and discusses the numerical results obtained using the simulated annealing algorithm. Particular attention is given to the effect of the algorithmic parameters and to the motivation for the segmented shift move. Although the simulation represents an abstract path-planning problem, it demonstrates that balancing terrain cost, path length, and curvature can produce non-trivial optimal paths.

Animations showing the evolution of the candidate path during the simulated annealing procedure are available in the accompanying GitHub repository [8].

We initially consider only the terrain cost term in the objective functional,

$$J(\gamma) = \int_{\gamma} w_c c(x, y) ds.$$

The numerical experiments in this section begin with this simplified formulation before progressively introducing the additional curvature and length terms required for the full model problem given in Equation (3.2). This allows the effect of each component of the objective functional to be investigated separately.

4.5.1 Performance of the Neighbourhood Moves

The first objective of these numerical experiments was to investigate the effectiveness of the neighbourhood construction and, in particular, to examine the motivation for using a cost-weighted segmented shift move. As discussed in Section 4.4.2, the basic shift move was limited by its ability to make substantial changes to the geometry of the path. Since only a single waypoint was moved at each iteration, the algorithm could require many iterations to move the path around a region of high terrain cost.

Figures 4.2 and 4.3 show a typical result obtained using the basic shift move. Over 10,000 iterations, the path cost decreased from approximately 156 to 150. The resulting path therefore showed only a relatively small improvement from the initial configuration, with no substantial change in the overall geometry of the path.

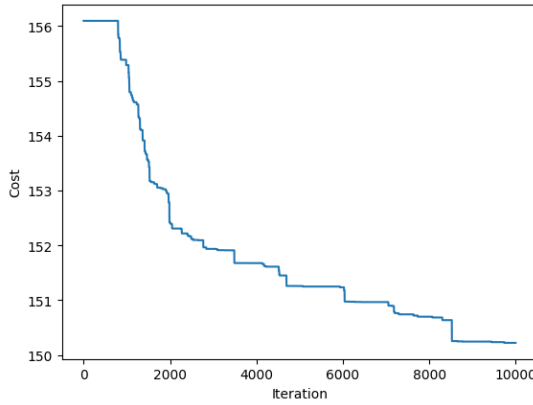


Figure 4.2: Evolution of the path cost using the basic shift move.

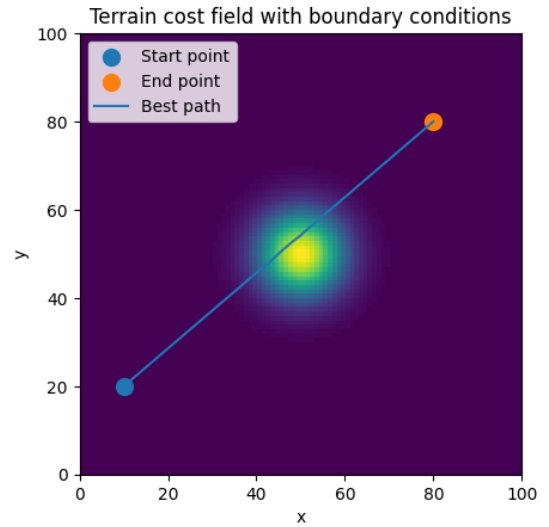


Figure 4.3: Final path obtained using simulated annealing with the basic shift move.

These observations motivated the development and experimentation of the cost-weighted segmented shift move described in Section 4.4.3. Keeping the remaining parameters the same, while allowing a greater number of iterations, the segmented move produced a substantially larger reduction in the objective cost. As shown in Figure 4.4, the cost decreased from approximately 156 to 10. The corresponding final path, shown in Figure 4.5, is able to move around the region of high terrain cost.

Recall that this experiment was conducted before the curvature and length penalties were introduced into the objective functional. Consequently, the resulting path is relatively long and contains sharp changes in direction. Nevertheless, these results demon-

strate the improved ability of the segmented shift move to produce large-scale changes in the path geometry and explore alternative regions of the solution space. An animation of this optimisation process is available in the accompanying GitHub repository [8].

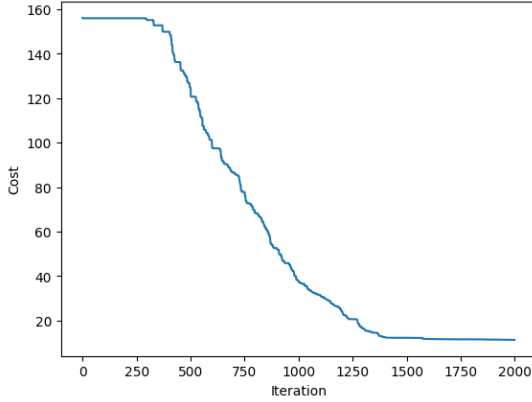


Figure 4.4: Evolution of the path cost using the segmented shift move.

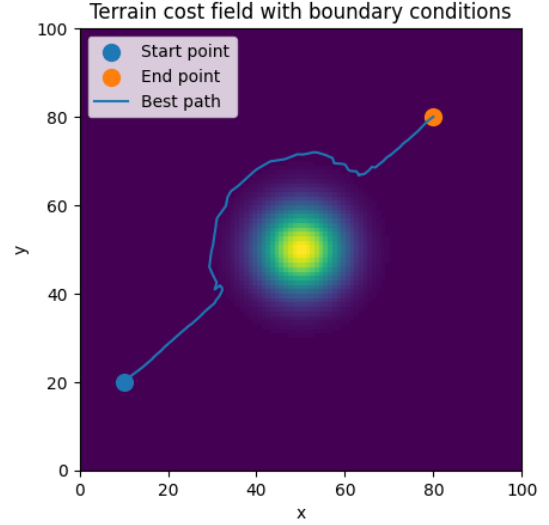


Figure 4.5: Final path obtained using simulated annealing with the segmented shift move.

4.5.2 Adding Curvature to the Model

Following the experiments with the neighbourhood moves, the objective functional was progressively extended towards the full model problem (3.2). The first additional term considered was the curvature penalty, giving

$$J(\gamma) = \int_{\gamma} [w_c c(x, y) + \lambda \kappa^2] ds.$$

The purpose of introducing this term was to investigate how penalising curvature affects the resulting path. In particular, the weighting parameter λ controls the relative importance of path smoothness compared with the terrain cost. Three values were considered, $\lambda \in \{1, 5, 20\}$, while keeping the other parameters fixed. The resulting costs are shown in Table 4.1.

Table 4.1: Effect of the curvature weighting parameter λ on the terrain and curvature costs.

	Initial	$\lambda = 1$	$\lambda = 5$	$\lambda = 20$
Terrain Cost	156.097	143.887	147.081	155.249
Curvature Cost	1.56×10^{-26}	0.278	0.271	0.0191
Total	156.097	144.165	147.352	155.268

For the initial straight-line path, the curvature cost is 1.56×10^{-26} , which is effectively zero as expected for a straight path. For $\lambda = 1$ and $\lambda = 5$, the total costs are similar, with values of 144.165 and 147.352, respectively. In both cases, the reduction in terrain cost is sufficiently large that the curvature contribution remains relatively small. For $\lambda = 20$, the curvature cost is reduced to approximately 0.0191, compared with 0.278 and 0.271 for $\lambda = 1$ and $\lambda = 5$, respectively. This corresponds to a reduction of over an order of magnitude in the curvature contribution. At the same time, the terrain cost increases to 155.249, indicating that the larger value of λ causes the optimisation to prioritise smoothness over avoiding the high-cost region.

The effect of this change can also be seen in the resulting paths in Figures 4.9, 4.10, and 4.11. For $\lambda = 1$ and $\lambda = 5$, the resulting paths pass through the higher-cost region while maintaining a relatively smooth trajectory. In contrast, $\lambda = 20$ places a much stronger penalty on curvature and produces a path which remains close to the initial straight-line configuration. This demonstrates that the curvature weighting is influencing the optimisation as intended, while also illustrating the trade-off between terrain cost and path smoothness.

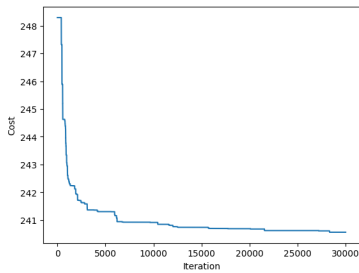


Figure 4.6: $\lambda = 1$

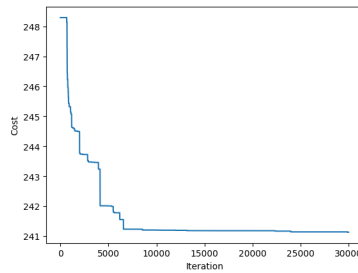


Figure 4.7: $\lambda = 5$

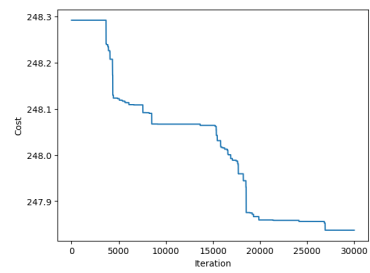


Figure 4.8: $\lambda = 20$

4.5.3 Adding Length to the Model

We now introduce the final term of the model, giving the complete objective functional considered in this chapter:

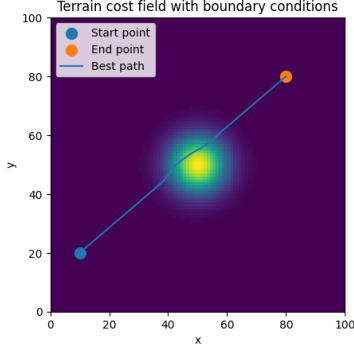


Figure 4.9: $\lambda = 1$

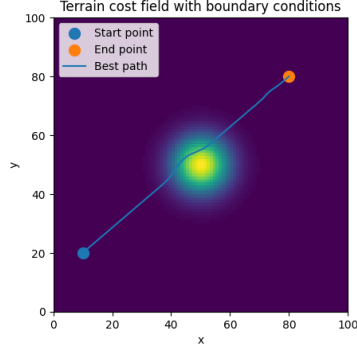


Figure 4.10: $\lambda = 5$

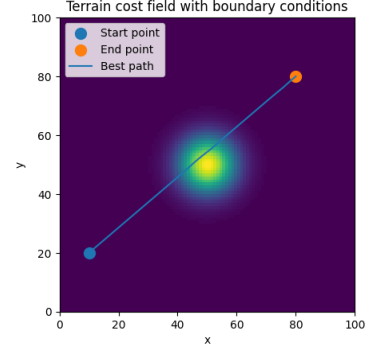


Figure 4.11: $\lambda = 20$

$$J(\gamma) = \int_{\gamma} [w_c c(x, y) + \lambda \kappa^2 + \mu] ds.$$

The three weighting parameters (w_c, λ, μ) determine the relative importance of terrain cost, curvature and path length, respectively. Increasing w_c encourages the path to avoid high-cost terrain, while increasing λ favours smoother paths and increasing μ penalises longer routes. The choice of these parameters therefore determines the balance between the three competing objectives.

Through experimentation, a weighting of

$$(w_c, \lambda, \mu) = (2, 20, 1)$$

was found to produce a satisfactory balance for the terrain considered in this study. The resulting optimisation is shown in Figure 4.12, while the final path is shown in Figure 4.13.

The resulting path demonstrates the trade-off between the three terms in the objective functional. Rather than completely avoiding the high-cost region, the path passes around the hill while maintaining a relatively smooth trajectory. The curvature penalty prevents the sharp changes in direction observed when only the terrain cost was considered, while the length penalty discourages the path from making an unnecessarily large detour.

Compared with the initial straight-line path, the final solution therefore represents a compromise between the competing objectives. The terrain cost is reduced by allowing the path to deviate from the straight line, while the curvature and length terms limit the extent of this deviation.

The weighting parameters used here are problem-dependent and were selected experimentally for the numerical example considered. They should therefore be interpreted as relative weighting parameters rather than universal physical constants. A more systematic

investigation of the parameter space could be used to determine how the resulting path changes as each component of the objective functional is prioritised.

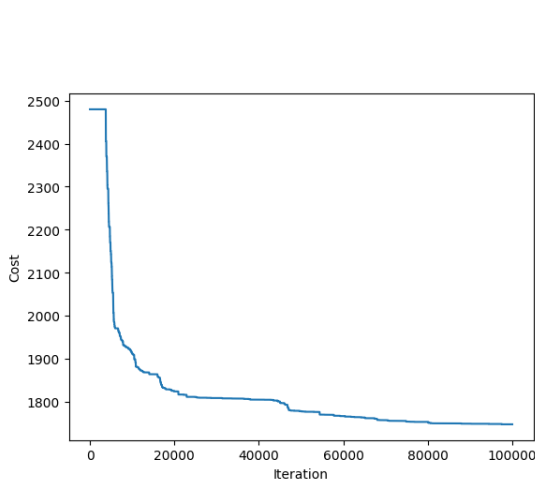


Figure 4.12: Evolution of the path cost.

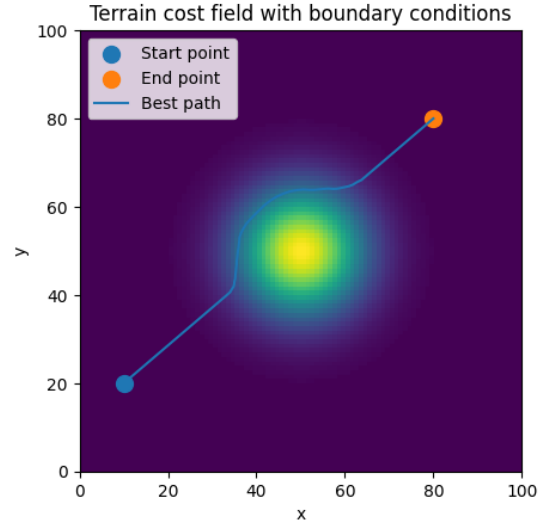


Figure 4.13: Final path obtained using simulated annealing.

4.5.4 Normalised Weights

The weighting parameters used in the previous experiments were selected experimentally and therefore introduced a degree of arbitrariness into the model. To investigate whether this could be reduced, the objective functional was reformulated by normalising each cost component with respect to its value for the initial path. The resulting functional was taken to be

$$J(\gamma) = \frac{w_c}{C_0} C(\gamma) + \frac{\lambda}{K_0} K(\gamma) + \frac{\mu}{L_0} L(\gamma),$$

where $C(\gamma)$, $K(\gamma)$ and $L(\gamma)$ denote the terrain, curvature and length costs, respectively, and C_0 , K_0 and L_0 are their corresponding values for the initial path. The weighting parameters are constrained such that

$$w_c + \lambda + \mu = 1.$$

The motivation for this formulation is to place the three cost components on a more comparable scale before applying their respective weights. However, the initial straight-line path has an almost zero curvature cost. Consequently, the value of K_0 is extremely

small, causing the normalised curvature term

$$\frac{\lambda}{K_0}K(\gamma),$$

to become disproportionately large and preventing a meaningful balance between the three terms. To address this, the initial path was first subjected to 50 random segmented shift moves. This produced a reference path with a larger curvature cost, allowing the three components to be normalised without the curvature term dominating the objective.

For the example considered, the resulting weights were

$$(w_c, \lambda, \mu) = (0.8348, 6.88 \times 10^{-29}, 0.0987),$$

with initial reference costs

$$C_0 = 1194.0065, \quad K_0 = 0.2609, \quad L_0 = 92.2243.$$

The resulting optimisation is shown in Figure 4.14, with the final path shown in Figure 4.15. The path provides a reasonable solution, although some irregular behaviour can be observed near the end point. One possible explanation is the interaction between the normalised cost weights and the cost-weighted selection of the segmented shift move, which influences where changes to the path are most likely to occur.

The normalisation therefore reduces some of the dependence on the absolute magnitude of the individual cost components, but it does not remove the need to select appropriate relative weights. In this example, the resulting path is less satisfactory than some of the solutions obtained using manually selected weights. Further investigation of the weighting scheme and its interaction with the neighbourhood construction would therefore be required.

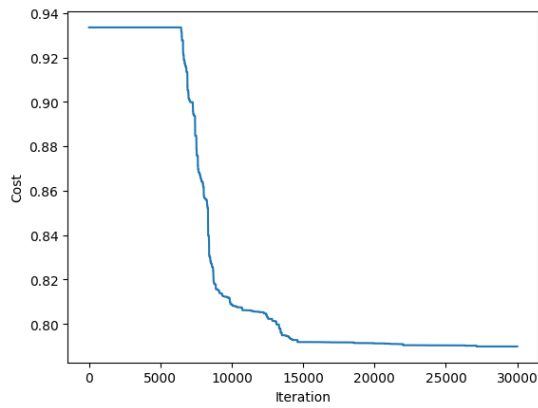


Figure 4.14: Evolution of the path cost using the segmented shift move.

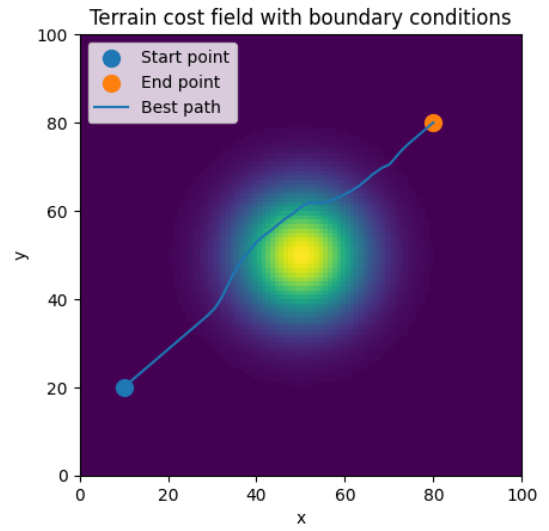


Figure 4.15: Final path obtained using simulated annealing with the segmented shift move.

Chapter 5

Discussion and Future Work

This dissertation has investigated an application of the Direct Method of the Calculus of Variations to a terrain-based path-planning problem. The core objective was to develop a variational model that considered terrain cost, path curvature and path length, and to investigate both the existence of minimisers and their numerical approximation.

The problem provides an example of the complementary roles of analytical and numerical methods in the Calculus of Variations. The Direct Method was used to establish conditions that minimiser exists, while numerical optimisation was used to find an approximate low-cost path.

5.1 Theoretical Conclusions

The proposed path-planning problem was formulated as a second-order variational problem. Applying the Classical Method to the corresponding functional leads to a fourth order Euler–Lagrange equation. Even for relatively simple choices of the terrain cost function, this still produces a nonlinear fourth-order boundary-value problem, making the an analytical solution difficult to find.

The Direct Method provides tool for showing existence of a minimiser. The existence result demonstrates that the infimum of the continuous functional is possible. However, the Direct Method does not provide an explicit path, and in this case, the results obtained do not establish uniqueness. Depending on the terrain cost field and the geometry of the problem, there may be multiple minimisers.

5.2 Numerical Conclusions

The numerical experiments presented in this chapter provide a controlled environment for investigating the proposed cost functional and optimisation method. The terrain model is intentionally simplified and should therefore be viewed as a numerical sandbox rather than a realistic railway or road planning model.

5.3 Limitations

The present work has several limitations. From a theoretical perspective, the existence results do not establish uniqueness or provide an explicit solution to the path-planning problem. Consequently, although the Direct Method establishes that a minimiser exists under the stated assumptions, it does not determine its precise form.

From a numerical perspective, the discretisation provides only an approximation to the continuous problem, with the curvature term in particular represented using a finite-difference substitute. Simulated annealing also provides no guarantee of finding a global minimiser, and the results may depend on the initial path, random choices and algorithm parameters.

The weighting parameters (w_c, λ, μ) and the annealing parameters were selected largely through experimentation. The cost-weighted segmented shift move also contains heuristic choices, meaning that different parameter values or neighbourhood constructions may produce different results. A more systematic investigation would therefore be required to establish the robustness of the numerical results.

Finally, the terrain considered is relatively simple. More complicated terrain, obstacles and higher spatial resolutions would increase the computational cost and provide a more demanding test of the method.

5.4 Further Work

A natural direction for further theoretical work would be to investigate stronger conditions under which additional properties of the minimisers could be established. In particular, it would be interesting to investigate whether further assumptions on the terrain cost function or the functional could provide information about uniqueness, regularity or the qualitative properties of minimising paths. An interesting direction for further investigation would also be to determine if a suitable choice of the terrain cost function c can lead to a reduction in the order of the Euler–Lagrange equation.

Numerically, a natural extension would be a systematic investigation of the parameter space to better understand the trade-off between terrain cost, curvature and path length. The numerical implementation could also be refactored using the lessons learned during its development, reducing the reliance on problem-specific heuristics.

The terrain model could be extended to include multiple terrain features and impassable regions, where the corresponding areas are removed from the admissible domain Ω . Introducing elevation would further extend the problem to three dimensions,

$$\gamma : [0, 1] \rightarrow \mathbb{R}^3,$$

allowing gradients, ramps and tunnelling to be considered.

A further long-term objective would be to apply the method to real-world geographical data. Mapping and terrain data could be converted into a discretised cost field, allowing the optimisation method to be tested on realistic route-planning problems.

5.5 Concluding Remarks

This dissertation has shown how the Direct Method of the Calculus of Variations can be applied to a path-planning problem for which the Euler–Lagrange equation is too difficult to solve.

The Direct Method provides an alternative analytical method by establishing conditions under which the continuous variational problem possesses at least one minimiser. Although this does not establish uniqueness or provide an explicit solution, it demonstrates that there is a minimising path to be approximated and provides a theoretical foundation for further numerical investigation.

The numerical experiments then provide an insight to what a optimal path may look like. They demonstrate how terrain, curvature and path length can interact to produce non-trivial low-cost paths. The paths produced by simulated annealing remain approximations and cannot be assumed to be global minimisers of the original continuous functional.

Overall, the analytical and numerical components of this dissertation serve complementary purposes. The Direct Method establishes that minimiser exists, while numerical optimisation provides a practical means of investigating the approximate form of the low-cost paths. Together, these approaches connect the existence of optimal paths with their numerical approximation and shows the value in combining variational analysis with numerical optimisation in the study of path-planning problems.

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